

How to succeed

- Read the Chapter ahead of lectures
- Come to class
- Start paper early
- Study groups
- Prepare exams
- Don't cheat, plagiarize, or otherwise participate in un-ethical behavior
- **Learn with Internet: Wikipedia, google(baidu)...**
- Ask questions
- Think skeptically

Research Paper Topics

- 3D Stereolithography/3D printing with polymers
- Plastic concrete
- Biocompatibility of silicones
- Teflon and fluoropolymers
- Piezoelectric polymers
- Plastic in the oceans
- Gas separation membranes
- polymeric foams
- The strongest plastic

Research Paper Topics

- Major catastrophe(s) due to a polymer
- Replacing ivory with plastic
- Biomimetic materials with polymers
- Plastic explosives and rocket fuels
- Bulletproof vest
- Plastics in Aerospace
- Polymer-based composites
- Hydrogels
- Super-glues
- Super-hydrophobic coatings/self-cleaning surfaces

Research Paper Topics

- Polymers from soybeans
- Furan based polymers from corn
- Bacterial and fungal attack on polymers
- Conducting polymers, new metallic materials
- Semiconducting polymers for PV
- Semiconducting polymers for OLED's
- Polymers for stealth
- Polymers for fire protection
- Smart polymers that change properties with external stimuli
- Reworkable, healable or removable polymers
-

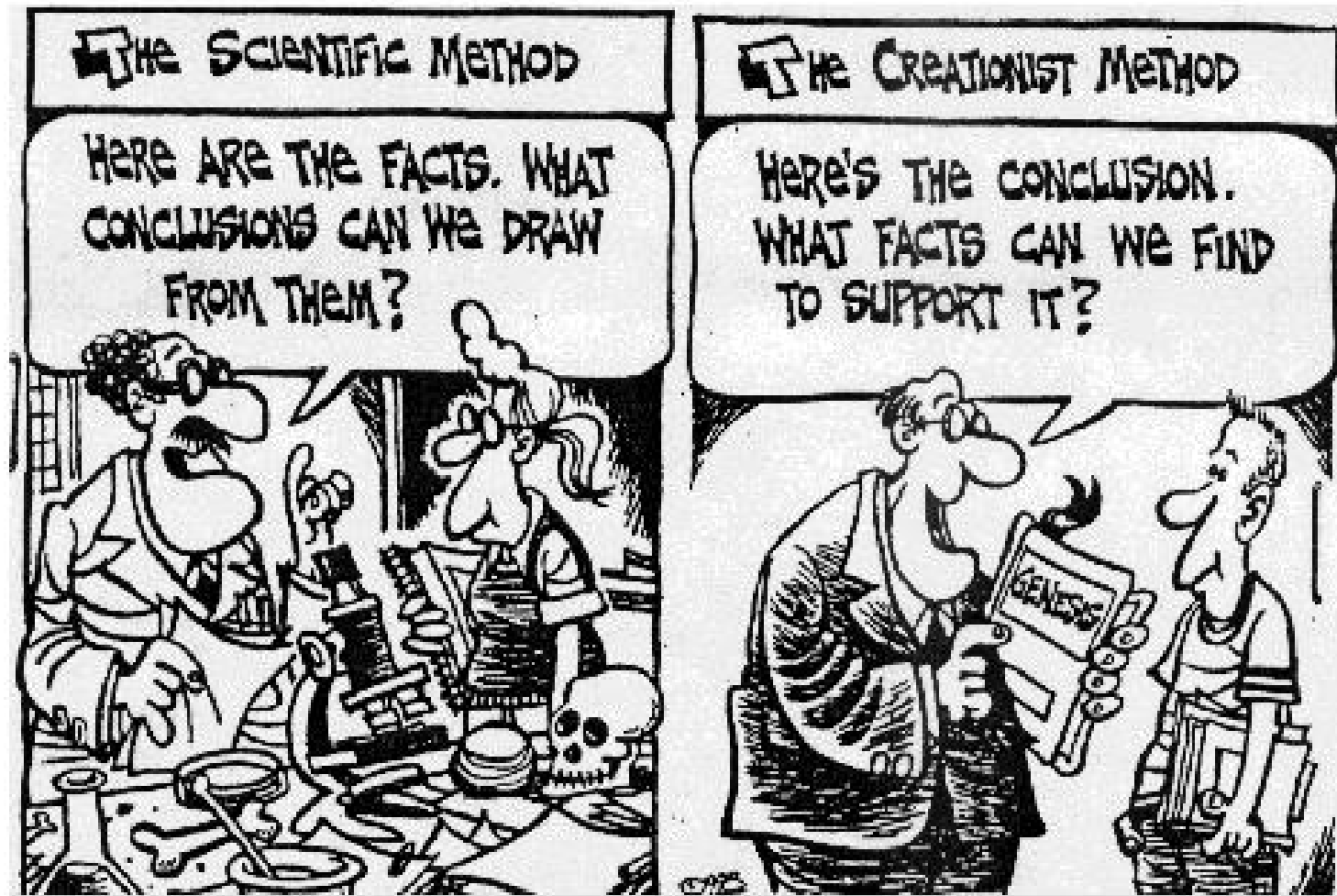
Research Paper

- *Review of literature topic I provide*
- *Petition* to present topic relating to research
- *ACS style bibliography, > 5 pages (double column)*
- *Keywords*
- *Suggest Endnote for editing references*
- Hard and electronic copy
- Literature search
- One draft due during semester

Keywords

- Derive from title
- Multiple keywords (3-5)
- Brain storm synonyms
- Without focus = too many unrelated hits

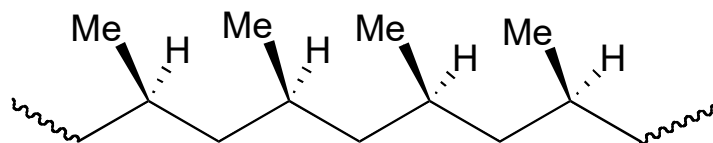
Scientific Research



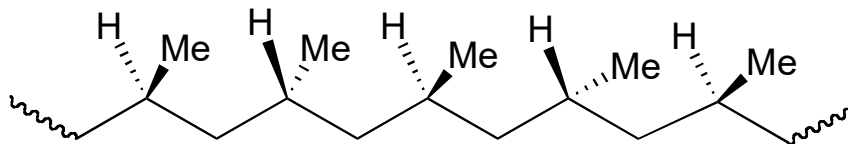
Polymer Tacticity

Polymer Tacticity: **Stereochemical** configuration

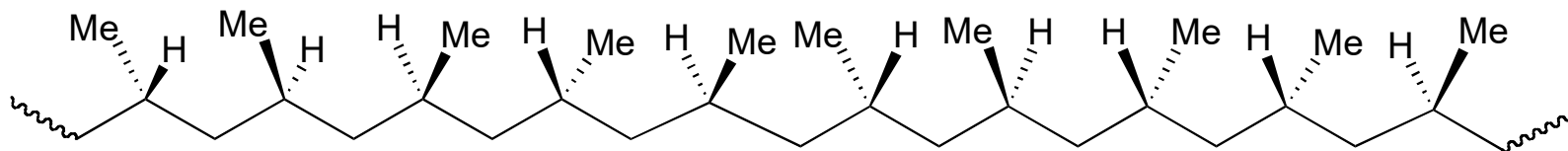
- typical for addition or chain growth polymers
- not for typical condensation or step growth polymers



isotactic



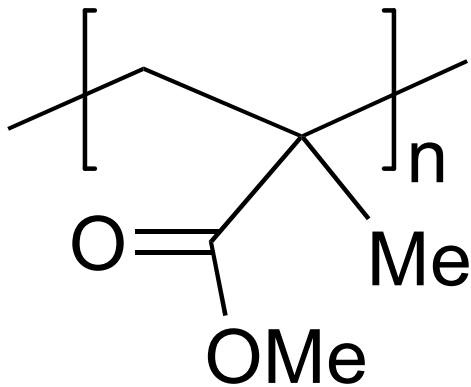
syndiotactic



atactic

Polymer Tacticity

Poly methyl methacrylate (PMMA)



Free radical - atactic

Anionic - isotactic

Why is this important?

- Tacticity affects the physical properties
 - Atactic polymers will generally be amorphous, soft, flexible materials
 - Isotactic and syndiotactic polymers will be more crystalline, thus harder and less flexible
- Polypropylene (PP) is a good example
 - Atactic PP is a low melting, gooey material
 - Isoatactic PP is high melting (176°), crystalline, tough material that is industrially useful
 - Syndiotactic PP has similar properties, but is very clear. It is harder to synthesize

Introduction to Polymer Materials

Lecture 2: Polymer Synthesis

SUN Dazhi

Polymerizations

Ring Opening Polymerization

Condensation Polymerization

Step Growth Polymerization

Metathesis Polymerization

Free Radical Polymerization

Addition Polymerization

Suspension Polymerization

Bulk Polymerization

Coordination Polymerization

Chain Growth Polymerization

Emulsion Polymerization

Plasma Polymerization

Anionic Polymerization

Cationic Polymerization

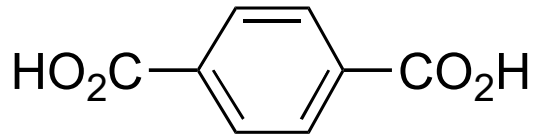
Gas Phase Polymerization

Solution Polymerization

Interfacial Polymerization

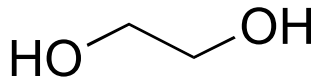
Step Growth Polymerization

Condensation

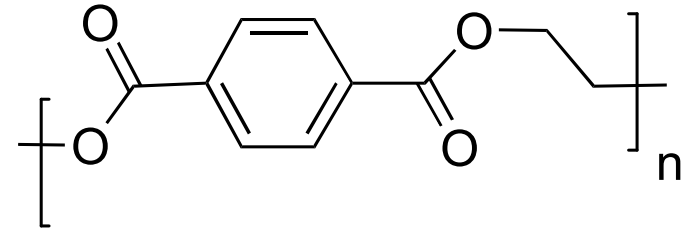


terephthalic acid

+



ethylene glycol



1:1 monomer ratio

**Poly(ethylene terephthalate): PET or PETE
Polyester**

**Two equivalents of water is lost or *condensed*
for each equivalent of monomers**

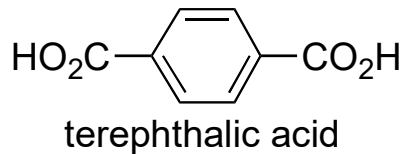


Dacron is a fiber

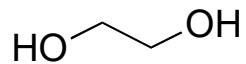


Step Growth Polymerization

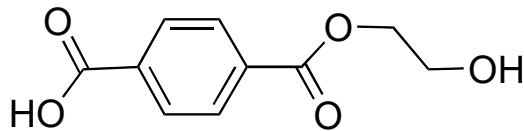
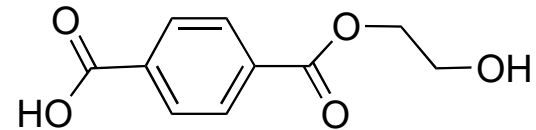
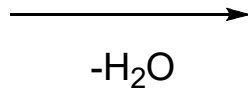
Condensation



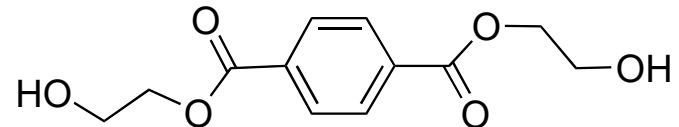
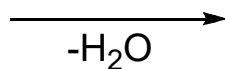
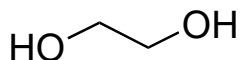
+



ethylene glycol



+



Biaxially stretched PET is “Mylar”

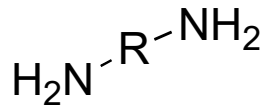
Step Growth Polymerization

- Epoxies
- Polyurethanes & ureas
- Nylon & polyesters
- Kevlar
- Polyaryl ethers (PEEK)
- Polysulphones
- Polyimides
- Polythiophenes & Photovoltaic polymers
- Polysulfides and polyphenyl ether

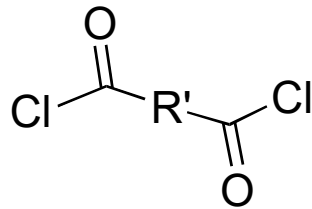
Mechanics of Step Growth

$$\text{Mole fraction Conversion} = 1 - [\text{COCl}]/[\text{COCl}]_0$$

AA

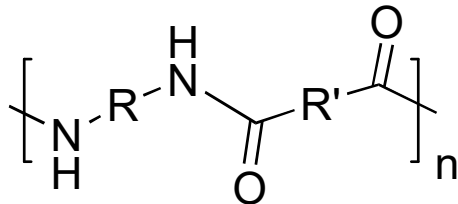


BB



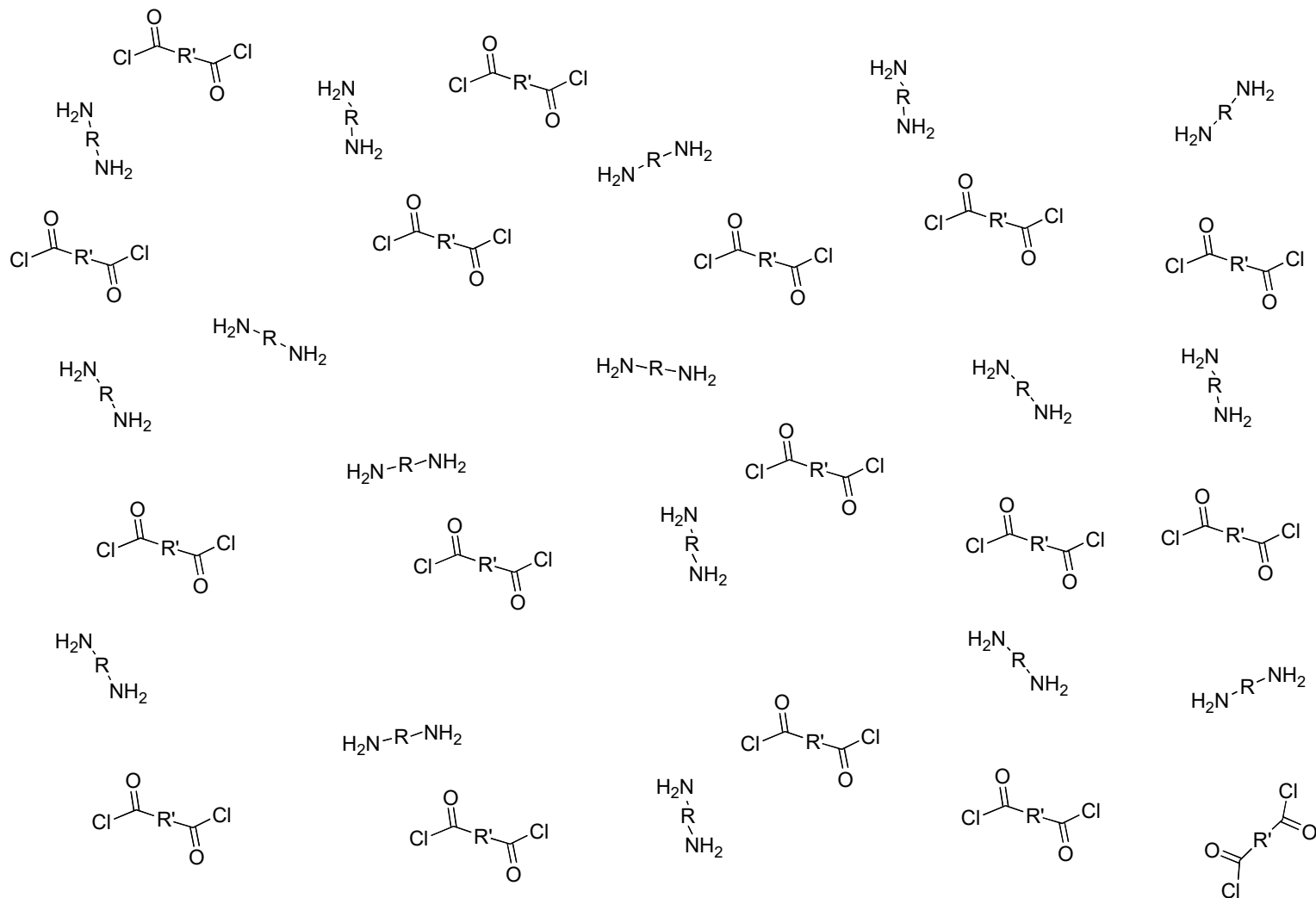
Each has functionality of 2

Can make two bonds



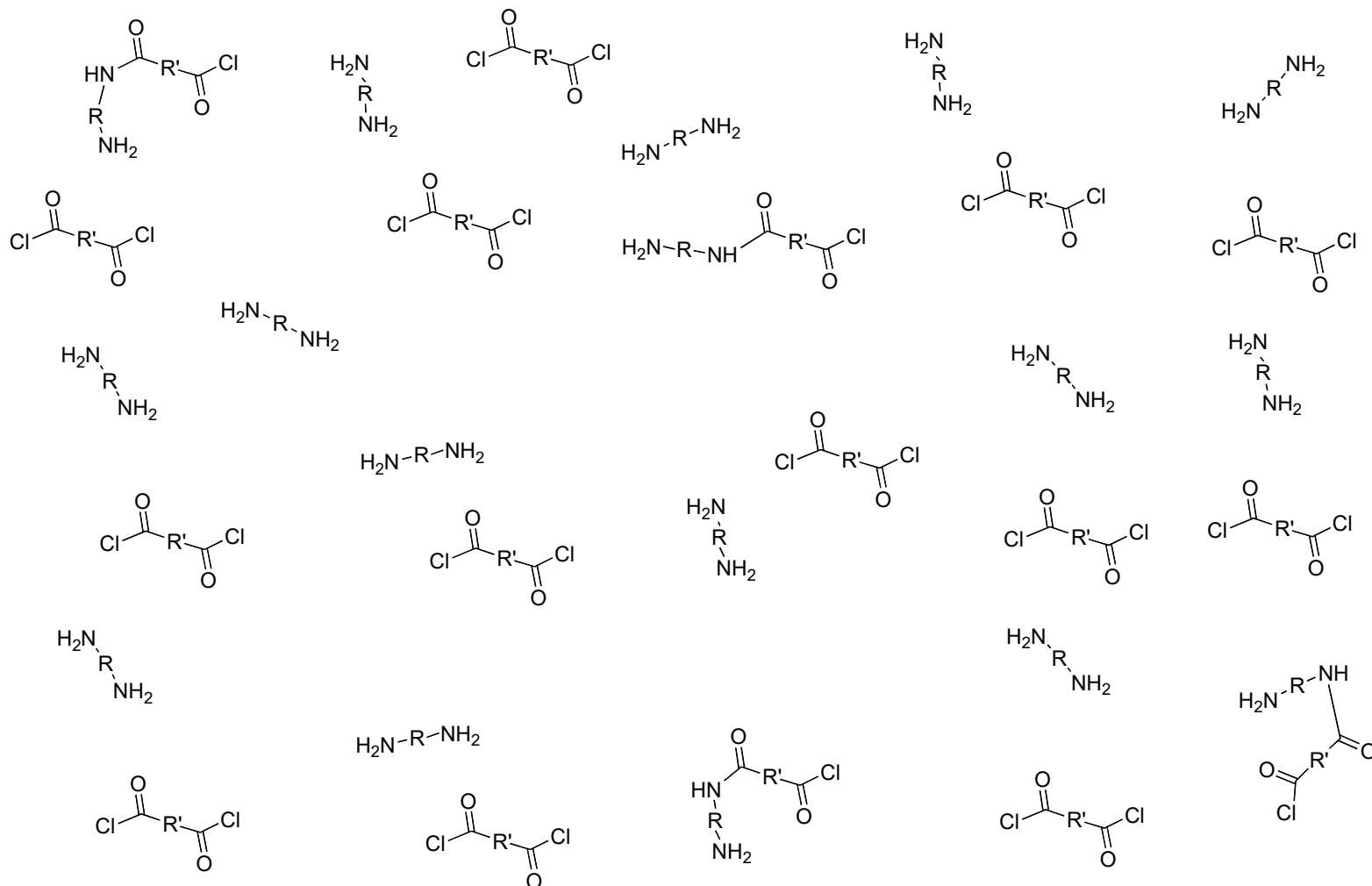
Linear, soluble Nylon polymer

Mechanisms of Step Growth: Monomers



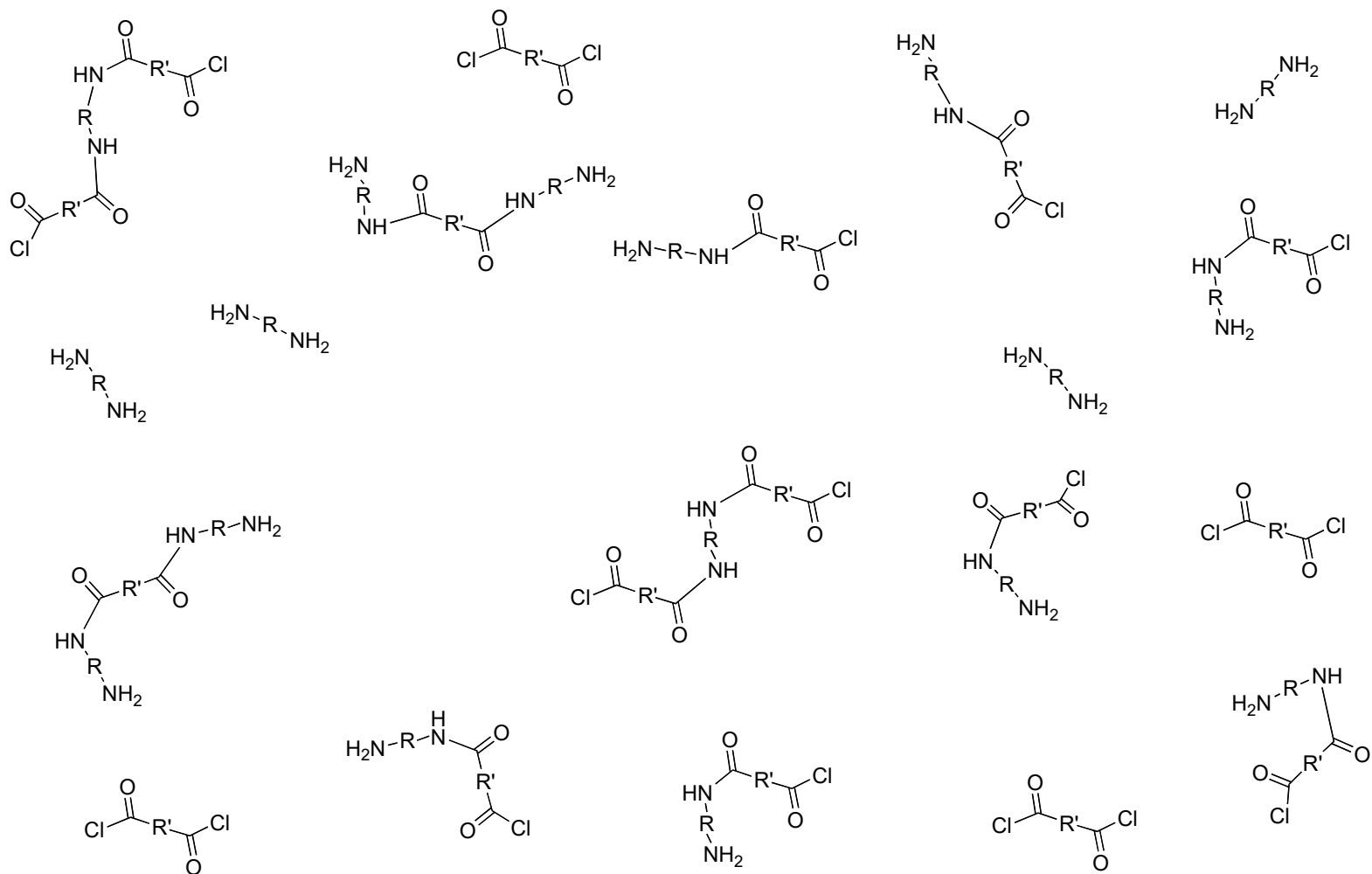
34 COCl groups; $p = 1 - [\text{COCl}]/[\text{COCl}]_0 = 0$ conversion

Mechanisms of Step Growth: M + Dimers



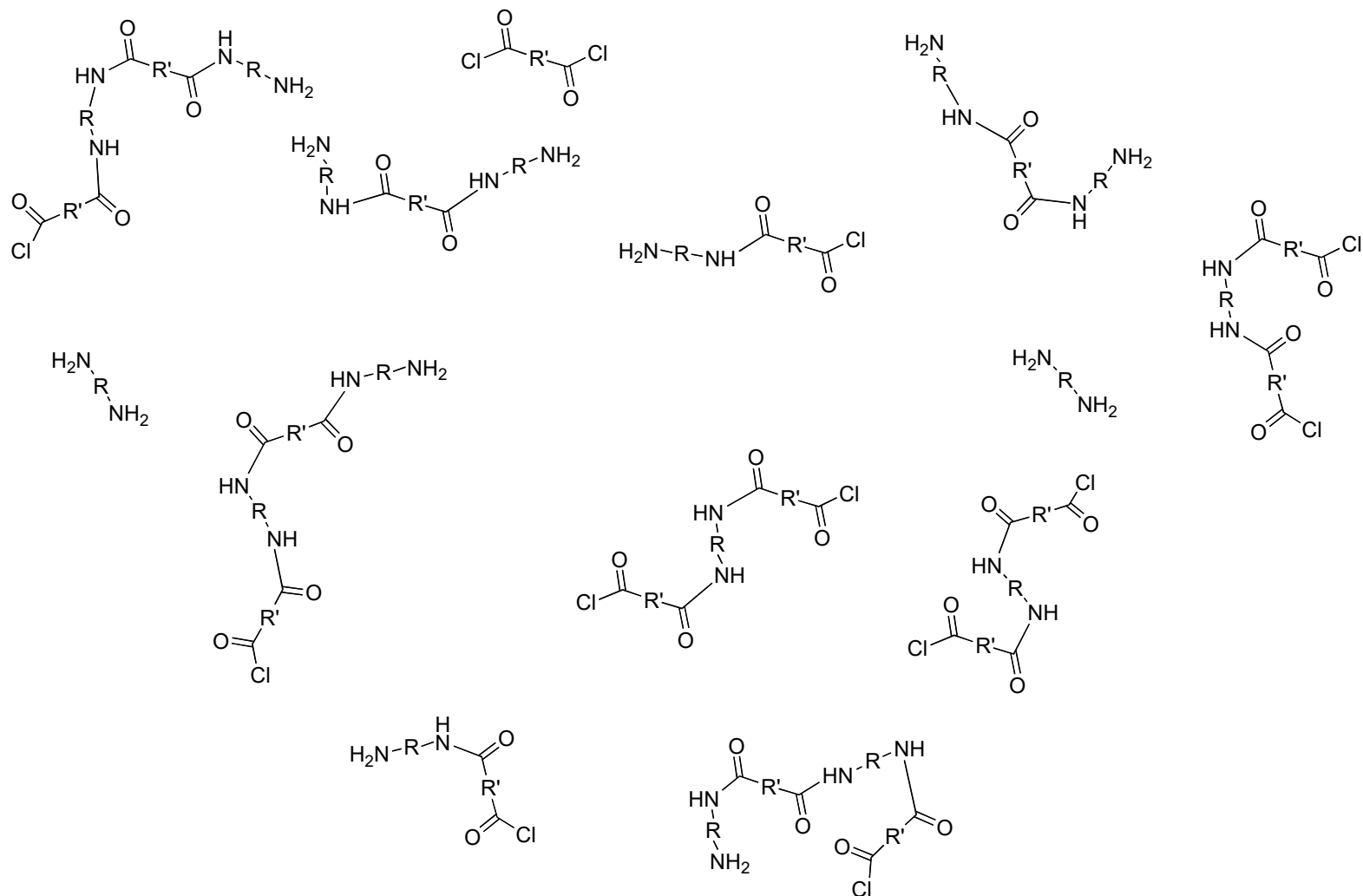
30 reactive groups $p = 1 - [\text{COCl}]/[\text{COCl}]_0 = 1 - 30/34 = 0.11$

Mechanisms of Step Growth: M + D + Trimers



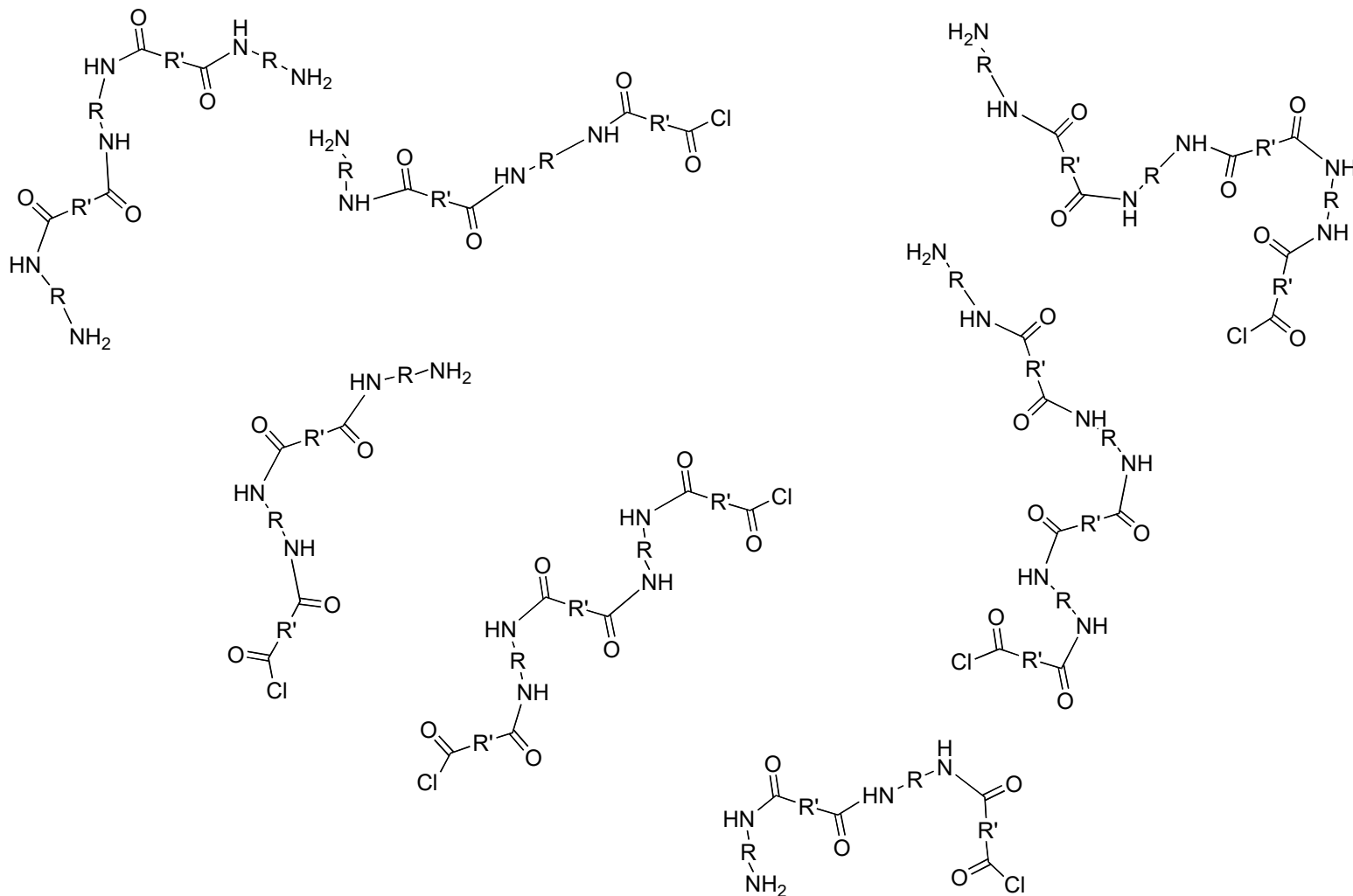
19 reactive groups $p = 1 - [\text{COCl}]/[\text{COCl}]_0 = 1 - 19/34 = 0.44$

Mechanisms of Step Growth: M + D + T + Tetramers



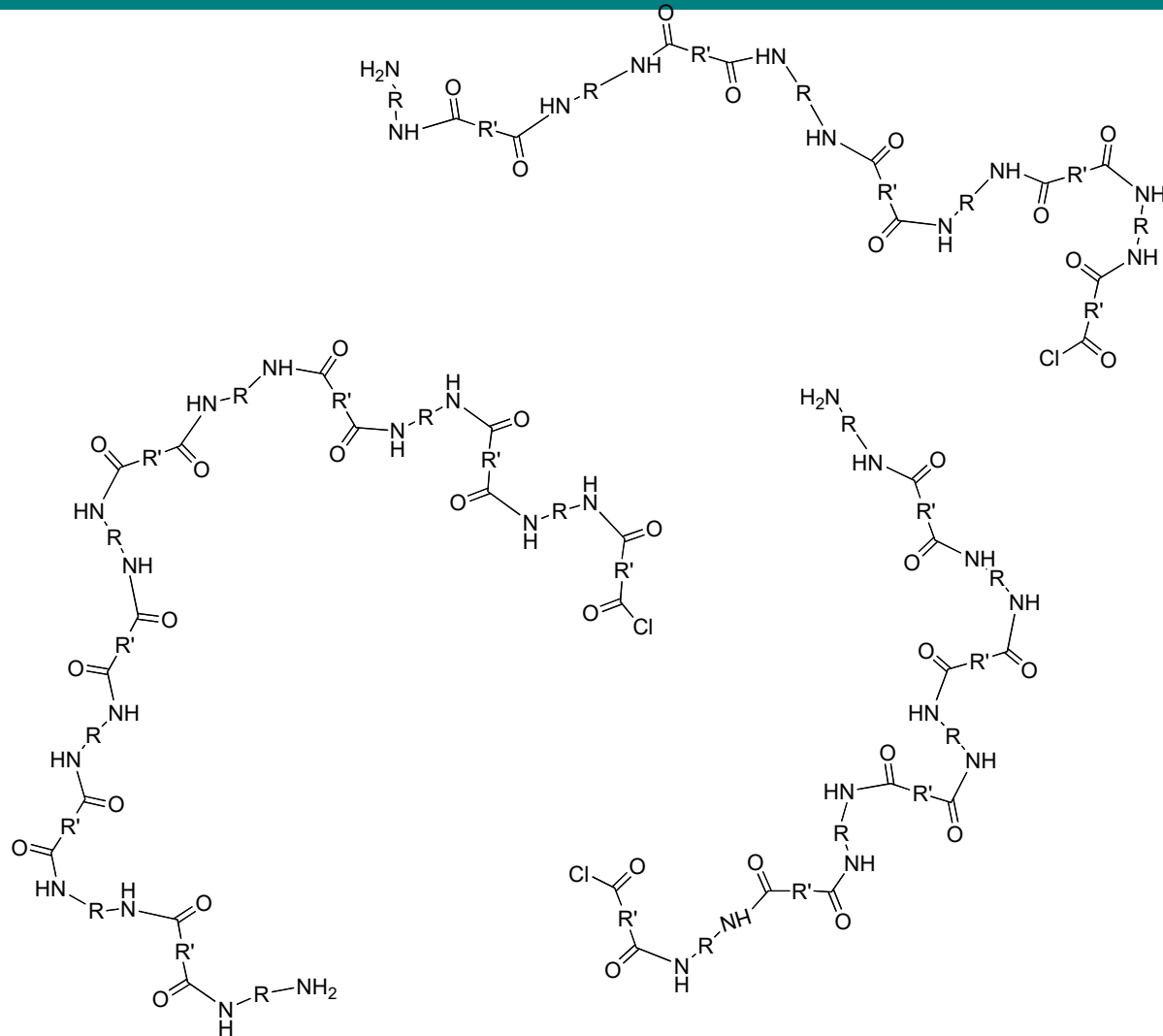
13 reactive groups $p = 1 - [\text{COCl}]/[\text{COCl}]_0 = 1 - 13/34 = 0.62$

Mechanisms of Step Growth: Higher



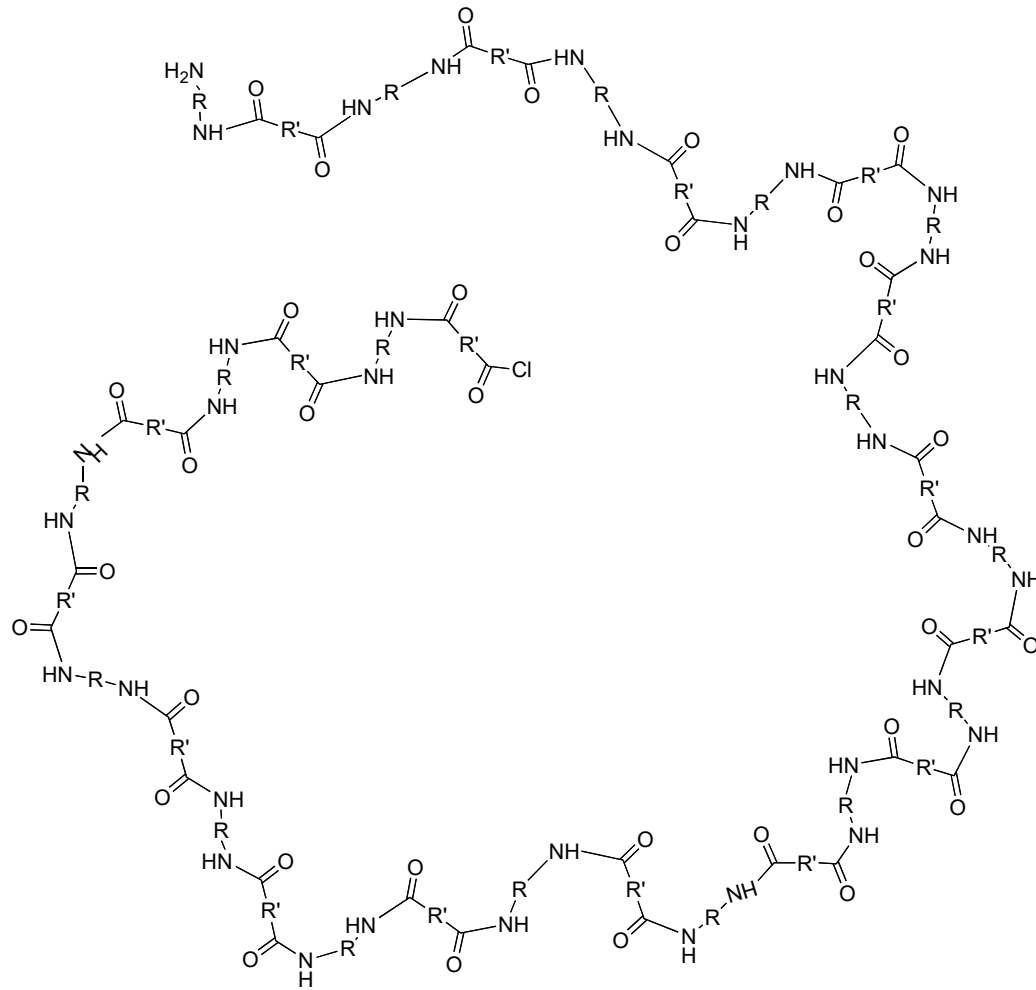
7 reactive groups $p = 1 - [\text{COCl}]/[\text{COCl}]_0 = 1 - 7/34 = 0.80$

Mechanisms of Step Growth: Higher



3 reactive groups $p = 1 - [\text{COCl}]/[\text{COCl}]_0 = 1 - 3/34 = 0.91$

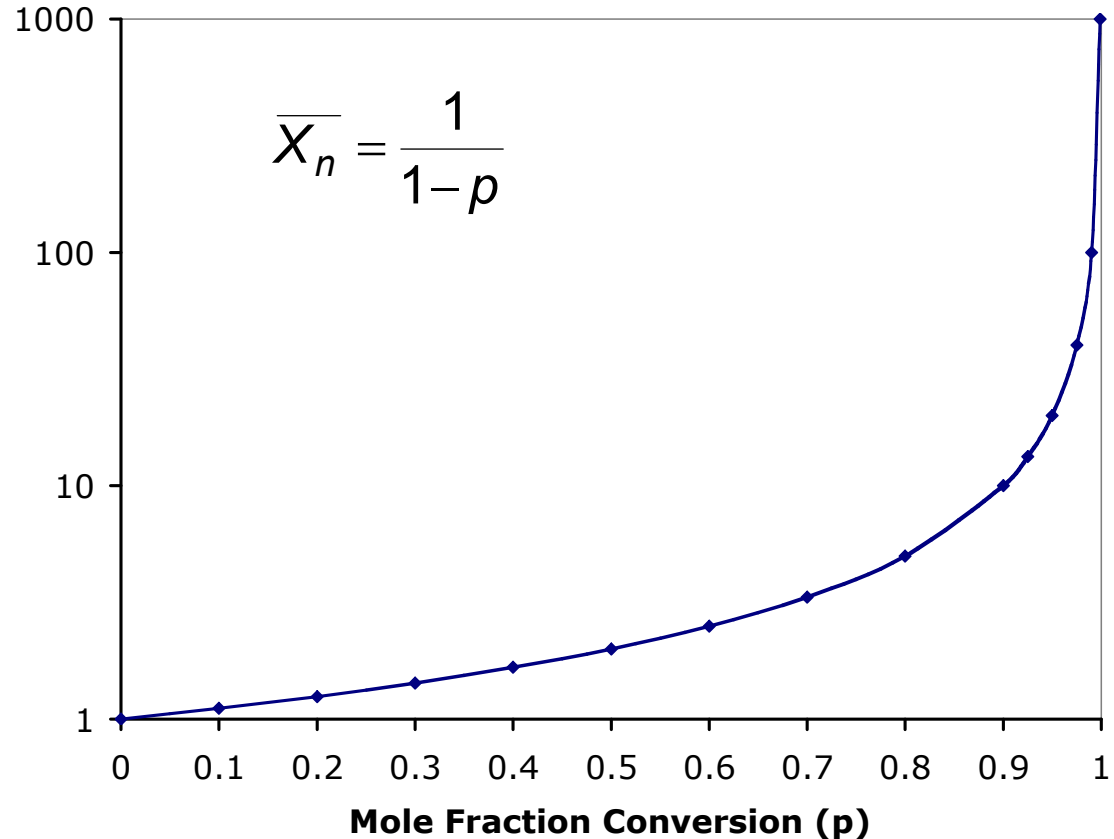
Mechanisms of Step Growth: Higher



1 reactive groups $p = 1 - [\text{COCl}]/[\text{COCl}]_0 = 1 - 1/34 = 0.97$

Step-Growth Polymerization

- Because high polymer does not form until the end of the reaction, high molecular weight polymer is not obtained unless high conversion of monomer is achieved.



$\bar{X}_n$ = Degree of polymerization
 p = mole fraction monomer conversion

Degree of Polymerization for step growth polymers

$$X = [\text{COCl}]_0 / [\text{COCl}] = 1 / (1-p)$$

$$\bar{X}_n = \frac{1}{(1-p)}$$

$$\bar{M}_n = \frac{M_0}{(1-p)}$$

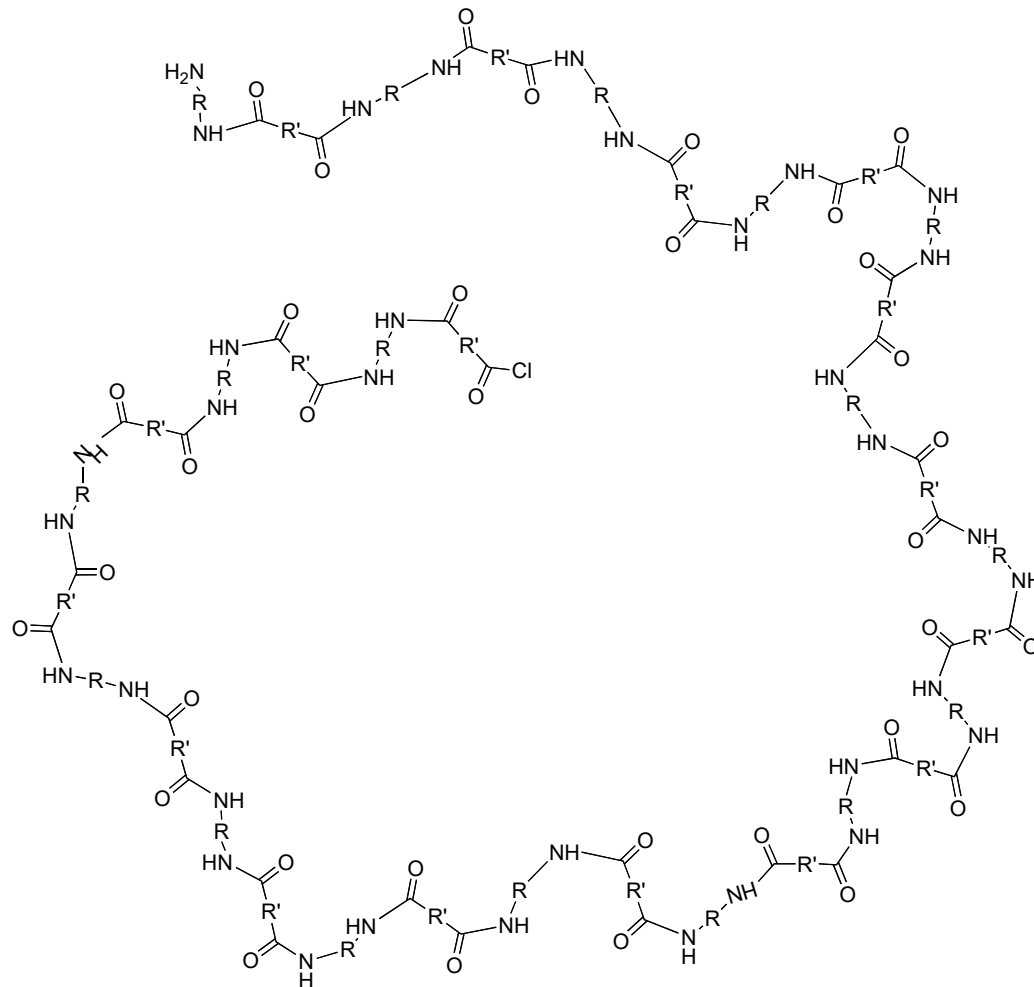
$$\bar{X}_w = \frac{(1+p)}{(1-p)}$$

$$\bar{M}_w = \frac{M_0 (1+p)}{(1-p)}$$

p = extent of reaction

M_0 = MW of monomer

Mechanisms of Step Growth: Higher



$$X \text{ or DP} = 1/(1-p) = 1/1-0.97 = 1/0.03 = 33$$

Impact of p on DP

Degree of Polymerization, D.P. = $N_o / N = 1 / (1 - p)$

Assuming perfect stoichiometry

if p =	DP =	DPmax= $(1 + r) / (1 - r)$ where r molar ratio of reactants
0.5	2	
0.7	3.3	
0.9	10	
0.95	20	if r = $[\text{Diacid}] / [\text{diol}] = 0.99$, then DPmax= 199
0.99	100	
0.999	1000	

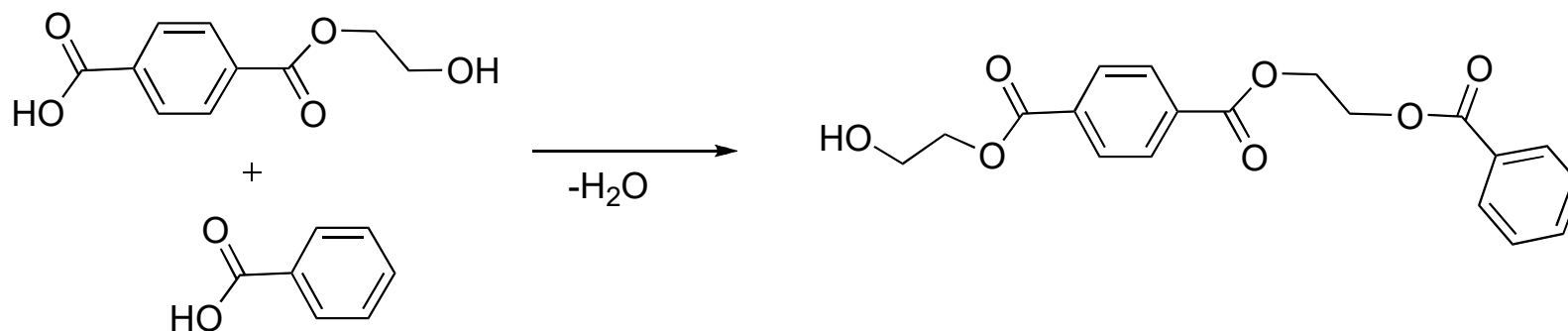
Problems in Achieving High D. P.

1. Non-equivalence of functional groups

- a. Monomer impurities
 - i. Inert impurities
 - ii. Monofunctional units terminate chain
- b. Loss of end groups by degradation
- c. Loss of end groups by side reactions with media
- d. Physical losses
- e. Non-equivalent reactivity
- f. Cyclization

2. Unfavorable Equilibrium Constant

Impurities



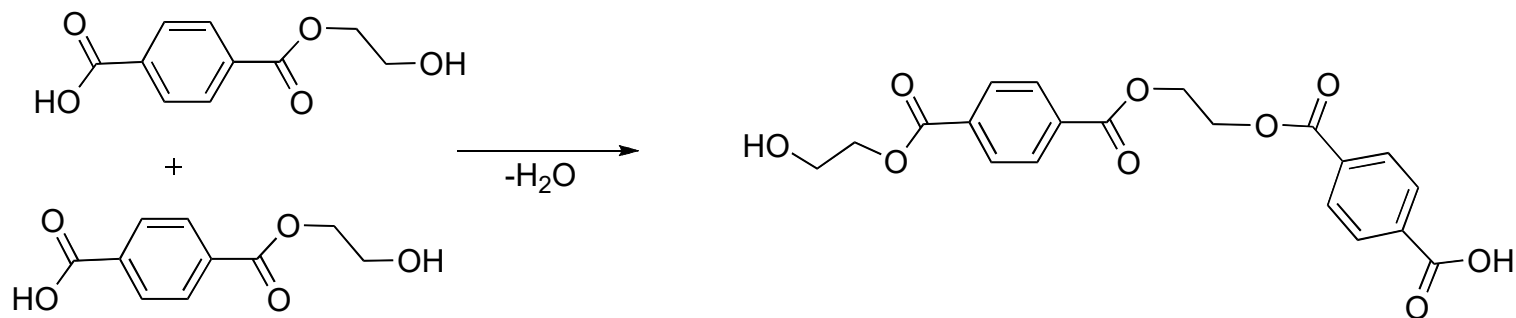
Impurities will kill growth and limit molecular weight

Problems in Achieving High D. P.

2. Unfavorable Equilibrium Constant

- Esters from Acids and alcohols $K_{eq} = 1-10$
- Amides from Acids and amines $K_{eq} = 10-1000$
- Amides or esters from acid chlorides, $K_{eq} > 10^4$

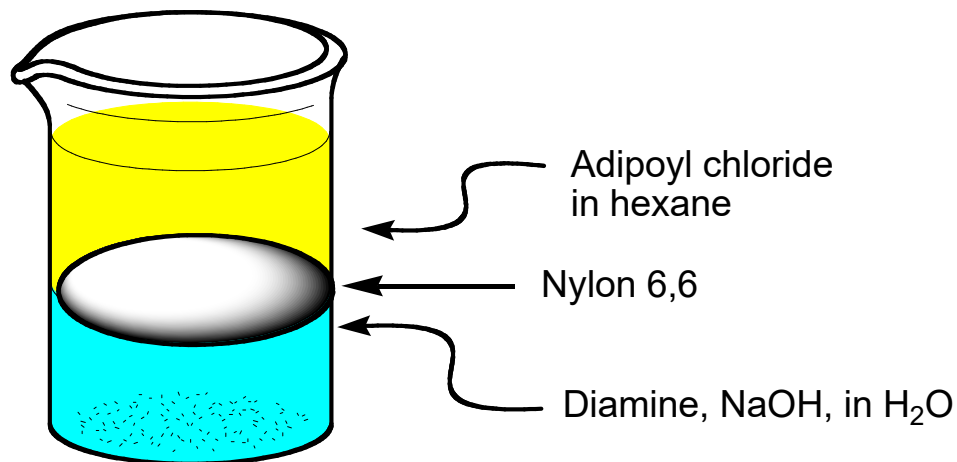
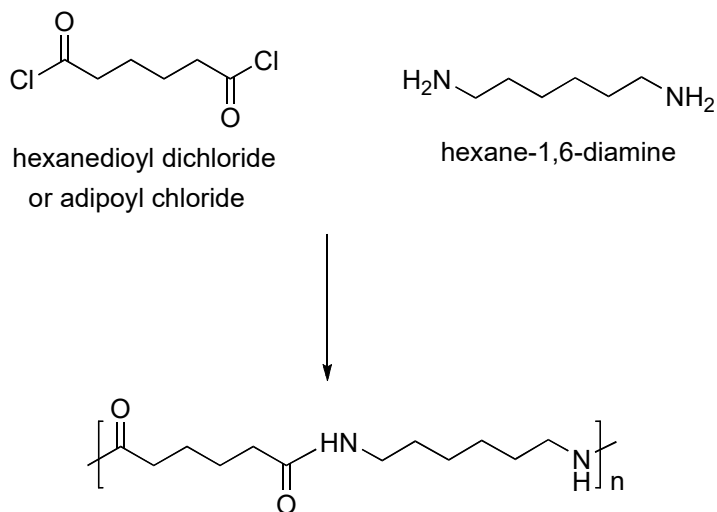
Reaction Equilibrium



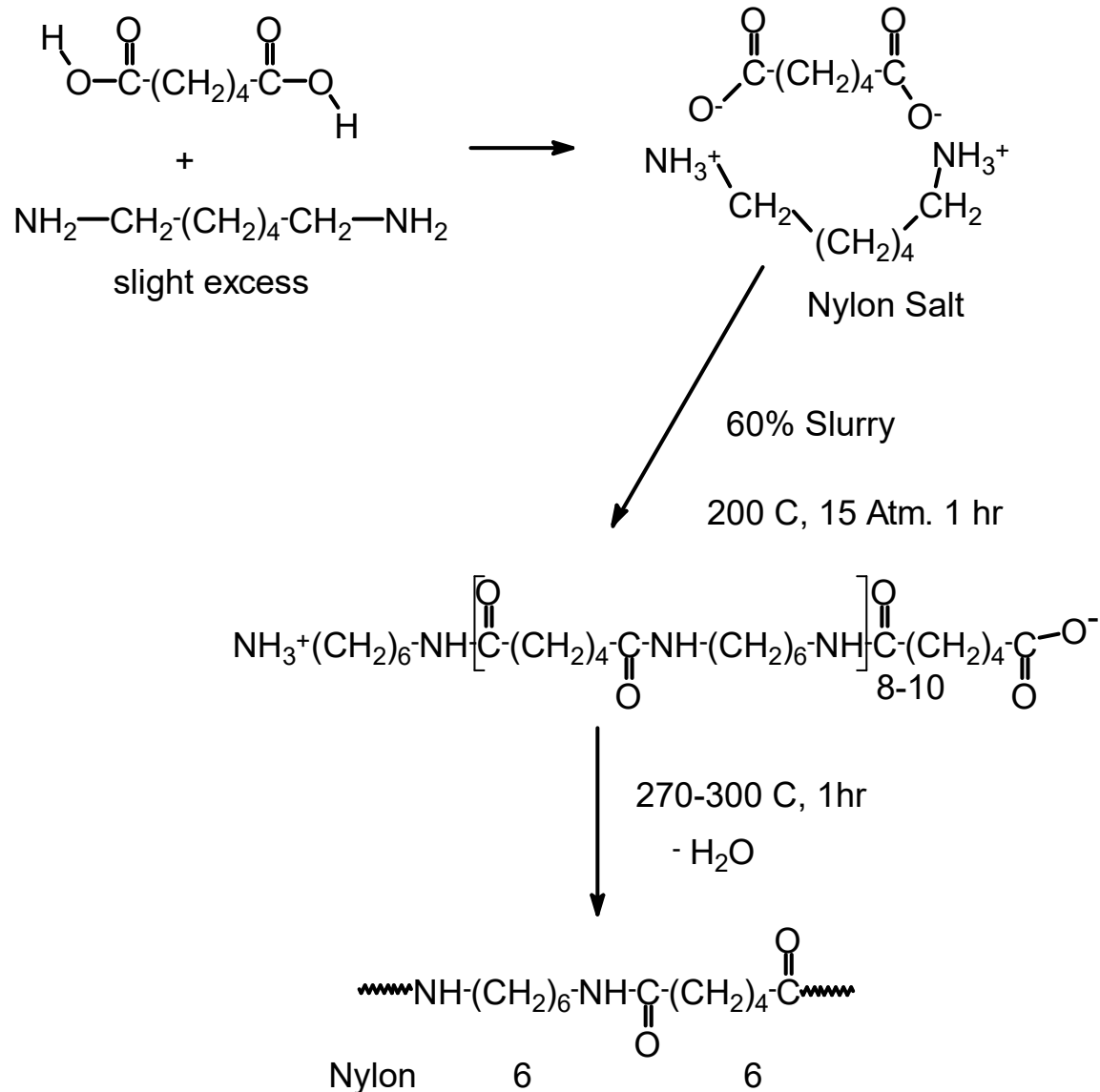
**Each reaction occurs at approximately the same rate.
Any monomer or growing oligomer can participate**

Interfacial Polymerization: Nylon Rope Trick

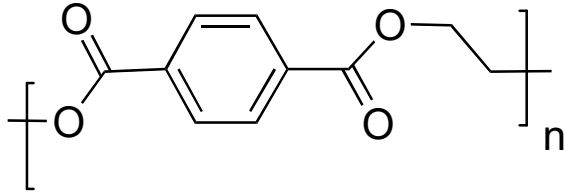
Driving Reactions forward with physics



Polyamides via Condensation -- Nylon 66

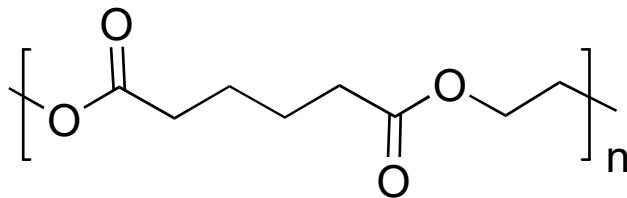


Biaxially stretched PET is “Mylar”



$$T_g = 70^{\circ} \text{C}$$

$$T_m = 265^{\circ} \text{C}$$



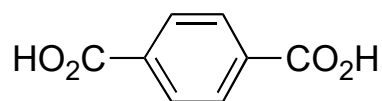
$$T_g < 0^{\circ} \text{C}$$

$$T_m = 50^{\circ} \text{C}$$

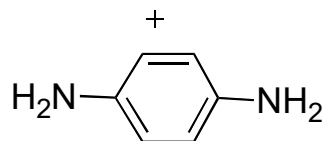


The Inventor

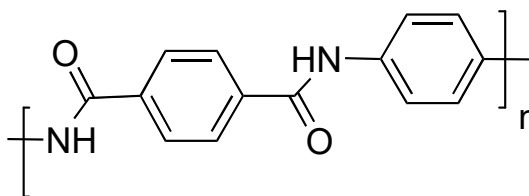
Polyamides



terephthalic acid



diaminobenzene



Kevlar



Stephanie Louise Kwolek
(DuPont)

Nylon Rope Trick

National Medal of Technology

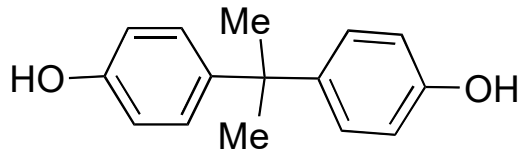
The Inventor



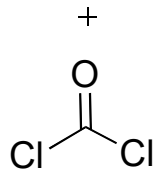
Nylon
Polyester
Polychloroprene (Neoprene)

Dr. Wallace Hume Caruthers
Head of DuPont Organic research Labs
50 patents

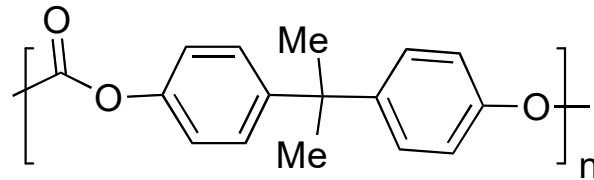
More Step Growth (Condensation) Polymers



Bisphenol A



phosgene

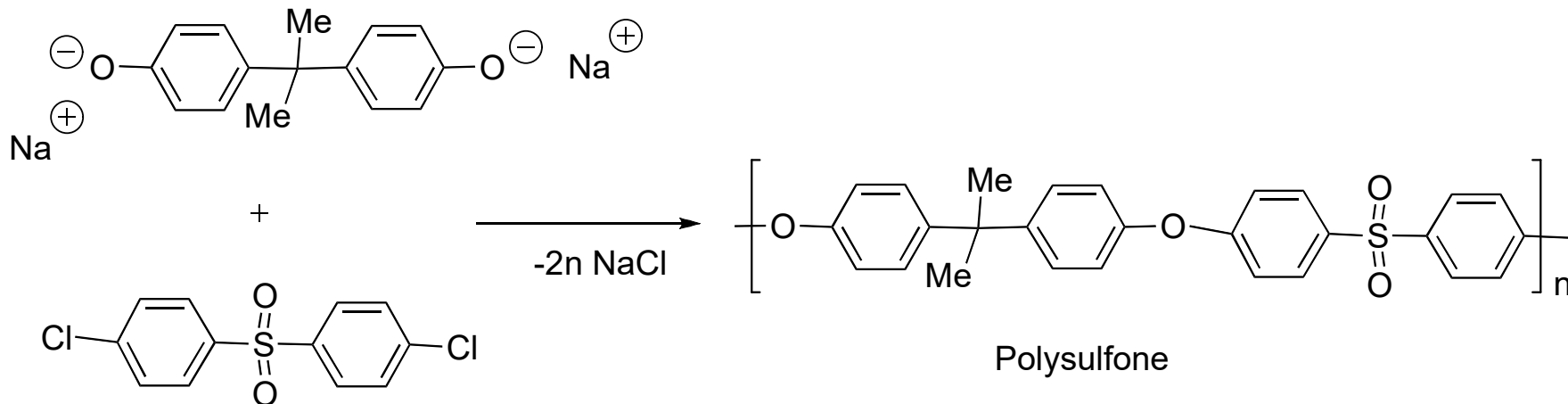


Polycarbonate
Lexan

$T_g = 150^\circ \text{C}$
 $T_m = 267^\circ \text{C}$

Two phase: interfacial polymerization

More Step Growth (Condensation) Polymers



$T_g = 200^\circ \text{C}$; Films pressed at 250°C

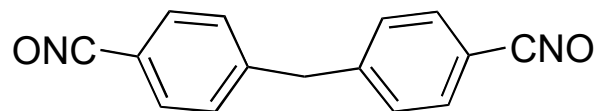
Use temperature $< 175^\circ \text{C}$

Stable in air to 500°C

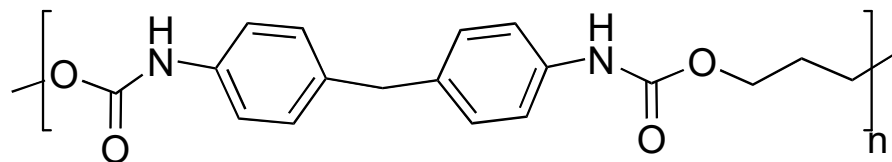
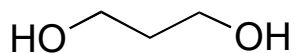
Self-extinguishing

More Step Growth (Condensation) Polymers

isocyanates



+



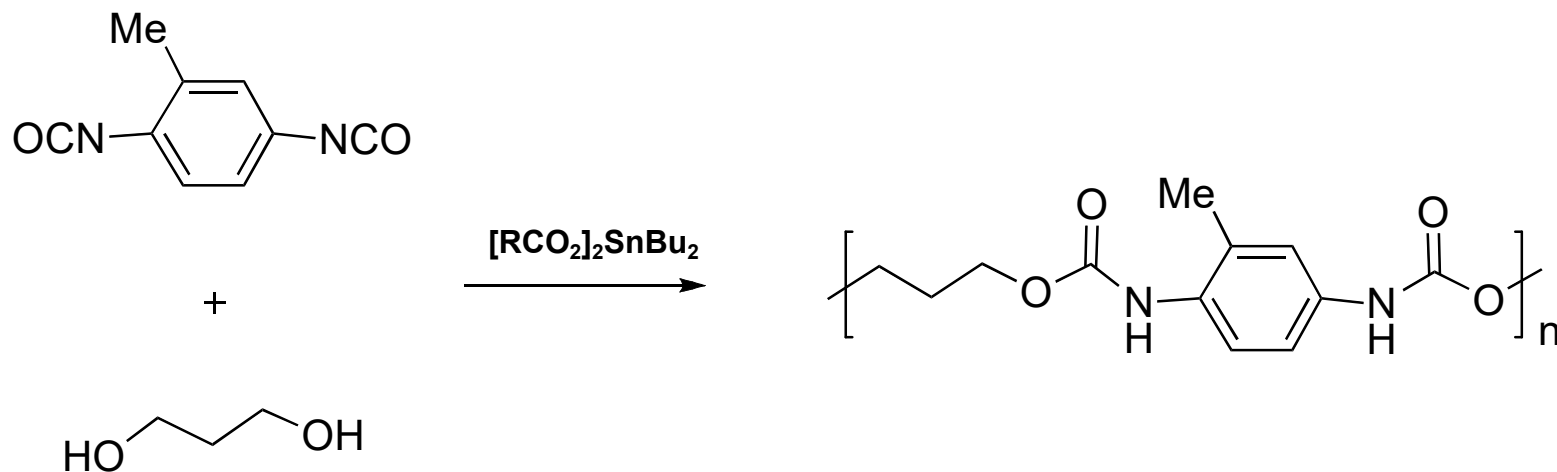
Polyurethane

Step Growth Polymers

- Polyesters, polyamides, engineering plastics such as polysulfones, polyetherether ketones (PEEK), polyurethanes.
- Condensation often occurs.
- High MW late in the reactions.

Step-Growth Non-Condensation Polymerization

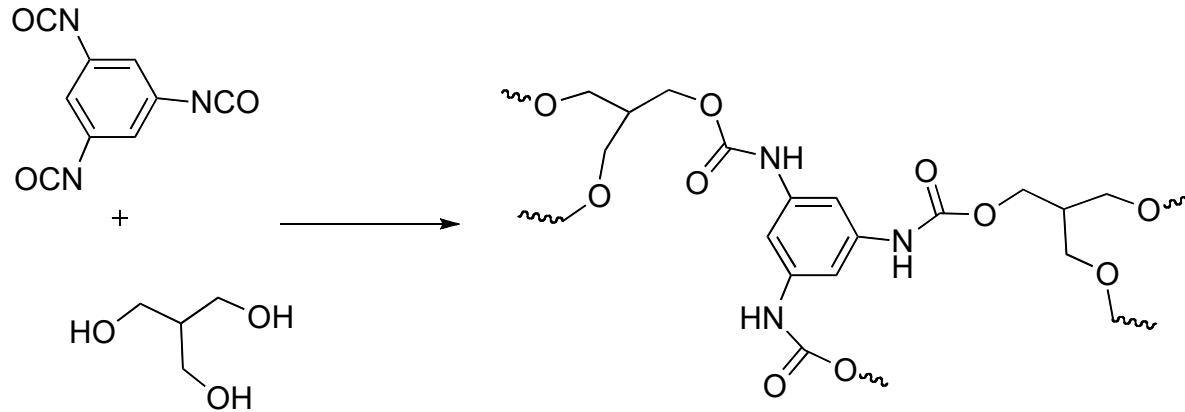
Polyurethanes



1,4-toluenediisocyanate + 1,3-propanediol

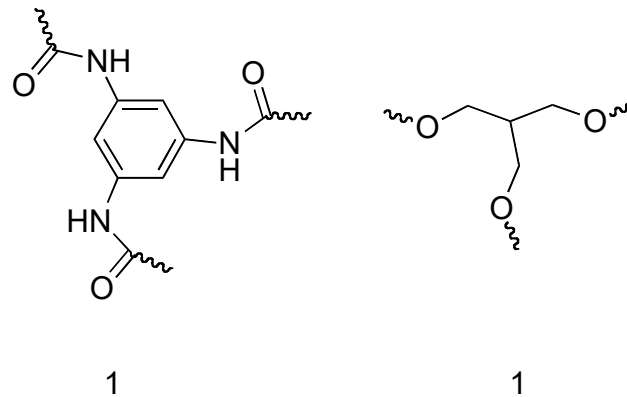
Functionalities > 2

Crosslinking into networks



Polyurethanes

$$f = 3$$



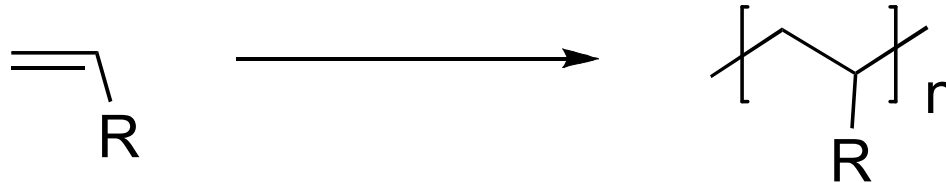
Thermosets

- Urethanes
- Epoxies
- Polyesters (2-stage)
- Formaldehyde-aromatic
- Melamine-formaldehyde

**Generally: Start as low viscosity liquids (low Mw)
And set or cure to form glassy “vitrified” solids.**

Chain Growth Polymerization

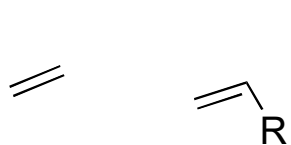
Addition Polymerization



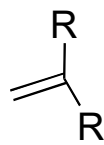
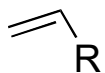
- 1) Catalyzed polymerization
 - free radical
 - cationic
 - anionic
 - coordination
- 2) Active group on end of polymer
- 3) MW increases more rapidly
- 4) Cheap & easier than step growth
- 5) Enthalpically favorable

Free Radical Polymerizations

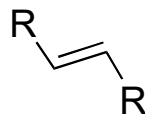
- Initiators (catalyst):
 - Thermal: azo compounds, peroxides,
 - Redox: persulfates
 - Photochemical: azo, peroxides, amine/ketone mixtures
- Monomers



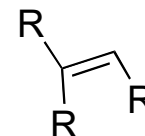
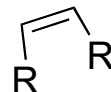
Polymerize fine



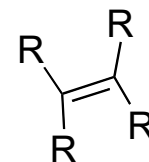
Usually
polymerize



Seldom polymerize



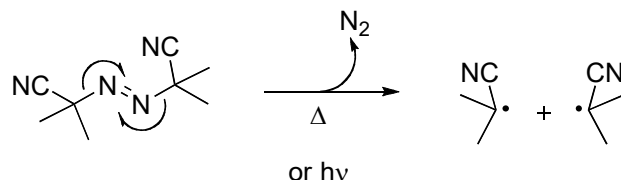
Almost never
polymerize



Almost never
polymerize

Free radical Mechanism

Initiation:

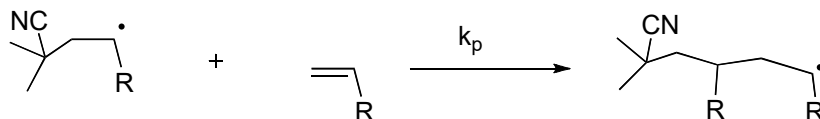


$$E_a = 140 - 160 \text{ kJ mol}^{-1}$$

$$K_d = 8 \times 10^{-5} \text{ s}^{-1}$$

$$t_{1/2} = 10 \text{ h at } 64^\circ \text{ C}$$

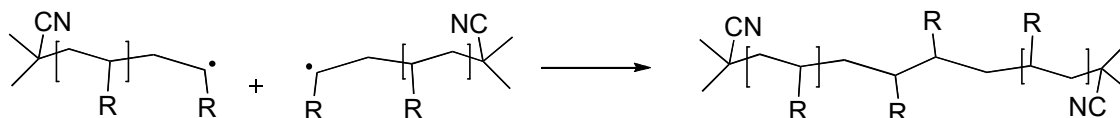
Propagation:



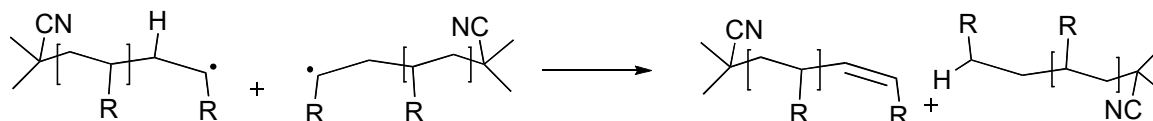
$$R_p = k_p [M\cdot][M]$$

$$k_p = 10^2 - 10^4 \text{ L/mol s}$$

Termination:

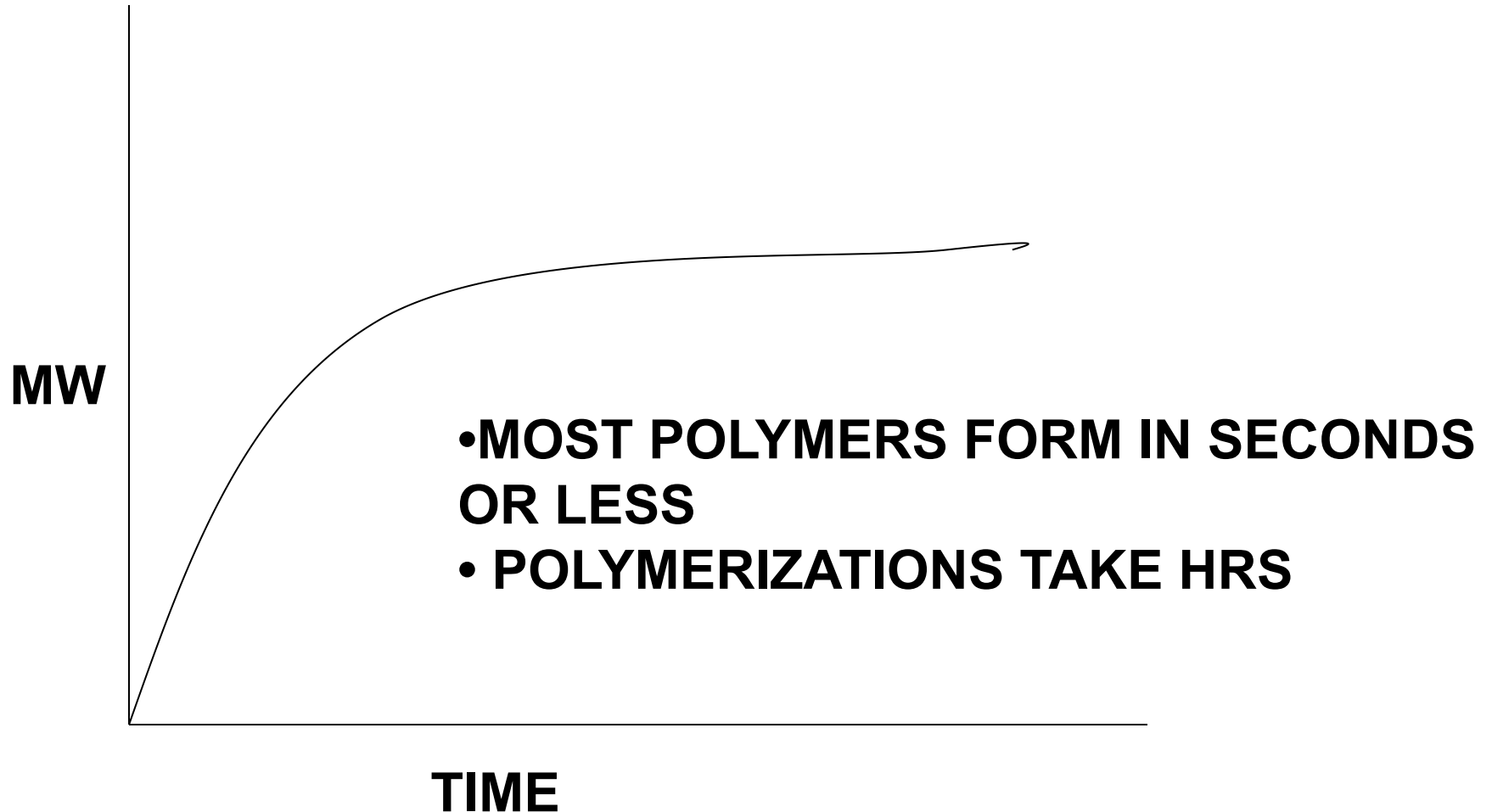


$$R_t = 2k_t [M\cdot]^2$$



$$k_t = 10^6 - 10^8 \text{ L/mol s}$$

Free Radical Polymerization Kinetics



Living Radical Polymerizations

Off-class learning (topics for Research Paper)

- 1) Atom Transfer Radical Polymerization (ATRP)
- 2) RAFT
- 3) TEMPO

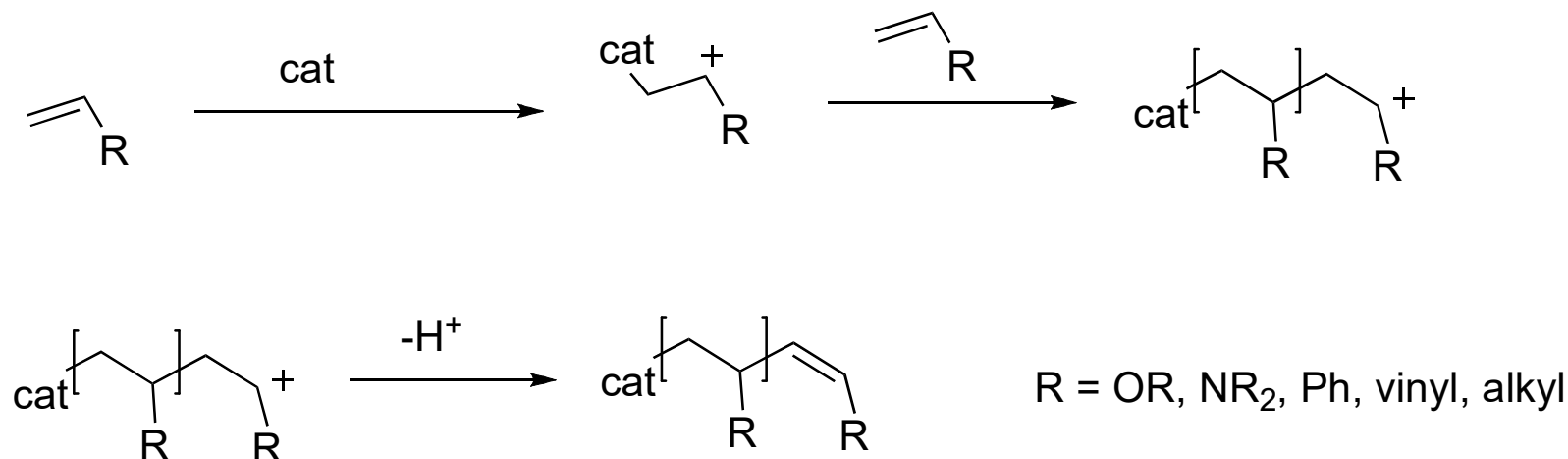
MW increases linearly with time

Narrow Mw distributions

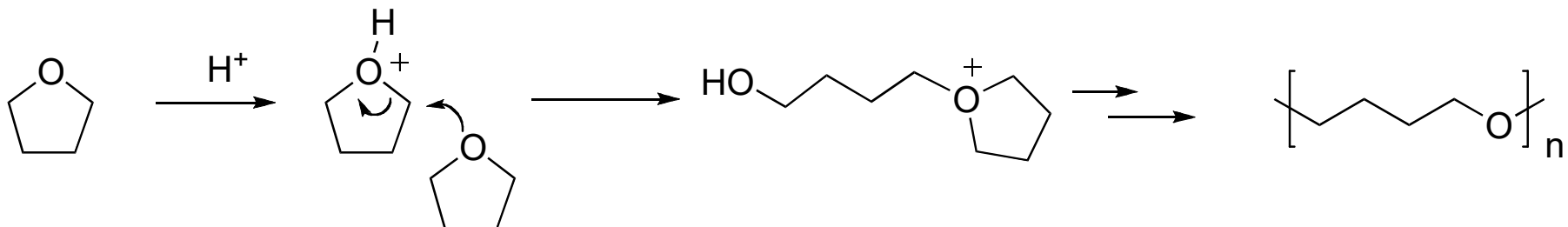
Block copolymers

Cationic Polymerizations

Vinyl polymerization

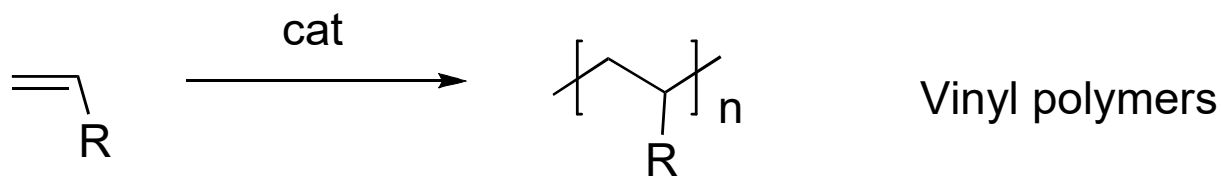


Ring opening polymerization

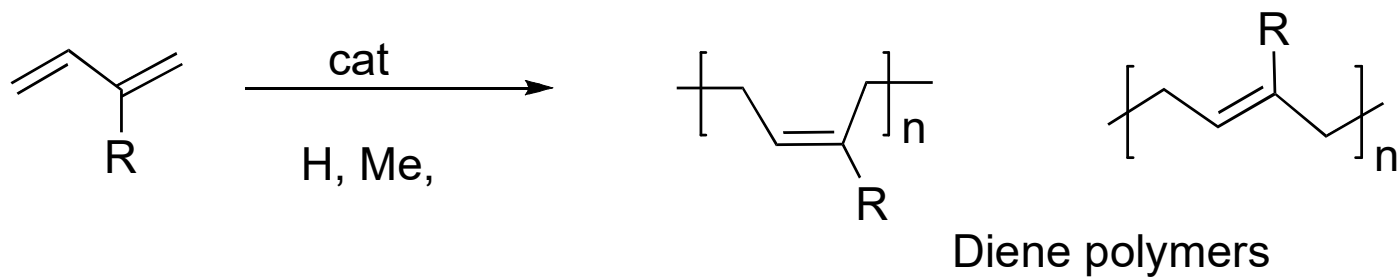


Anionic Polymerizations

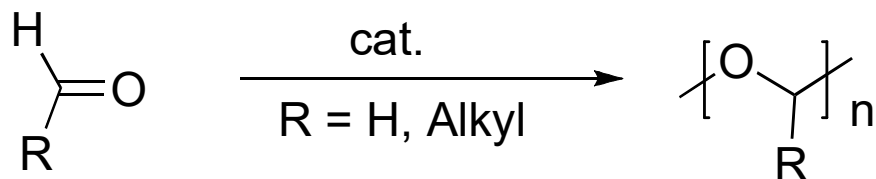
cat = Alkyl or aryl Lithium, sodium naphalide, alkyl Grignard, some alkoxides



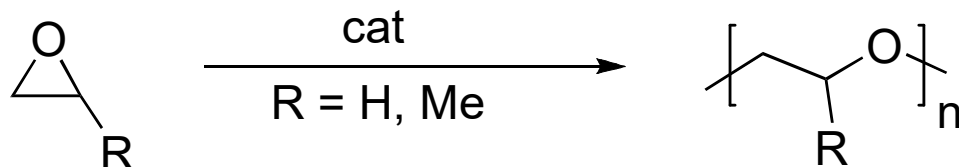
R = Ph, vinyl, CO₂R, CN



Anionic Polymerizations

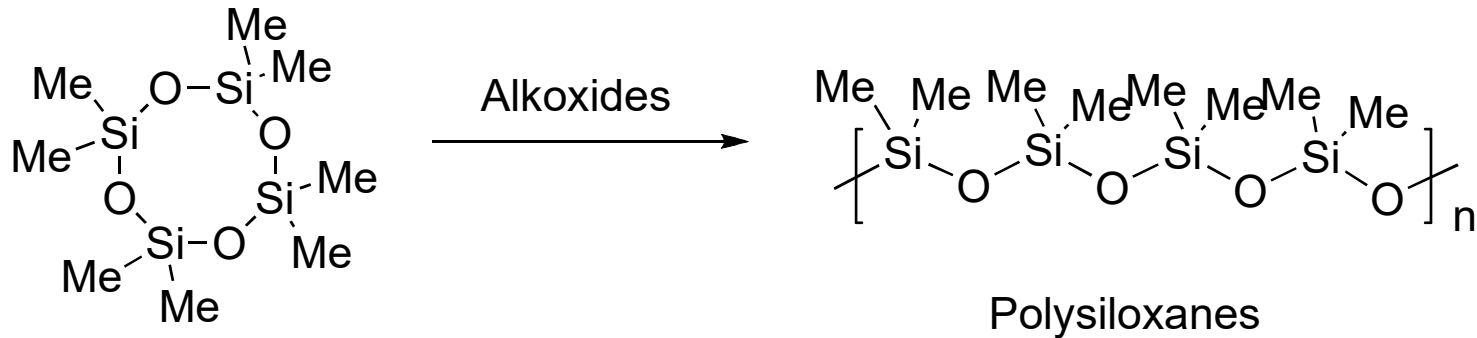


Polyacetals or carbonyls



Poly ethers

Anionic Polymerizations

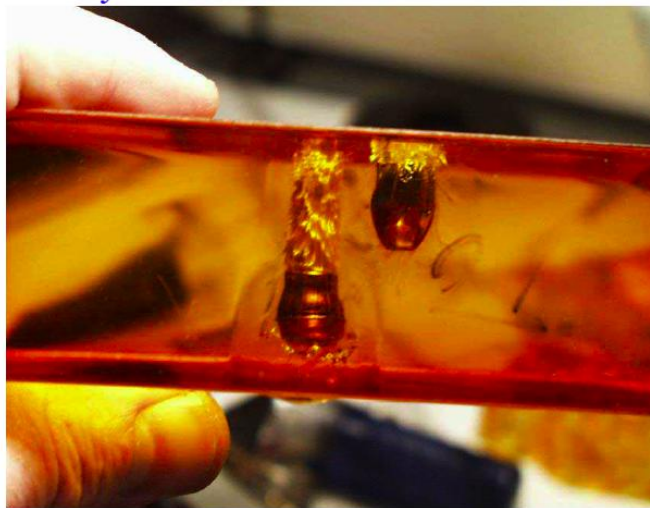


Coordination Polymerizations

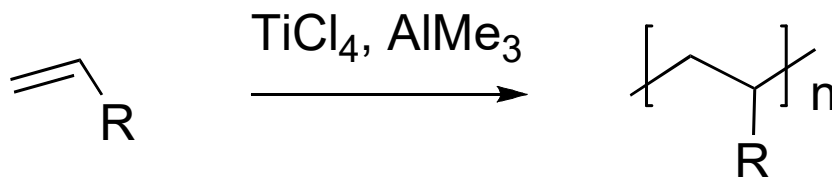
Transition Metal Mediated Polymerizations

- Ziegler Natta polymerizations
- Metathesis polymerization **Off-Class**
- Insertion polymerizations **Off-Class**

PolyDCPD-9mm Ballistic Protection

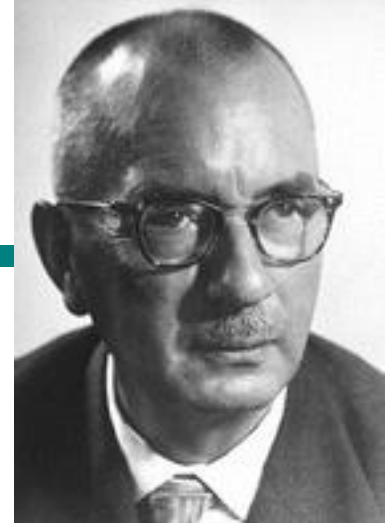


Ziegler Natta Polymerizations



- ZN are heterogeneous; solid catalysts
- Catalytic polymerizations
- Polypropylene and HDPE
- Highly productive: 10^6 g polymer/gram catalyst-hour
- Stereochemical control:

iso or syndiotactic polymers

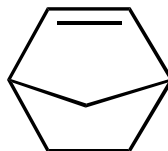
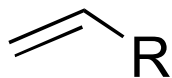


Karl Ziegler (1898-1973)

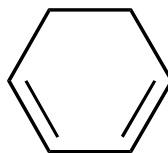
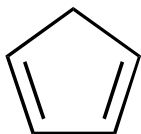
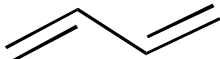
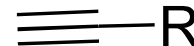


Giulio Natta (1903-1979)

Ziegler Natta Monomers



α -olefins
styrenes



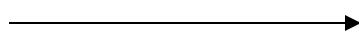
R = alkyl, aryl

Not compatible with heteroatoms (O,N,S,etc)

Complex Coordination Catalysts

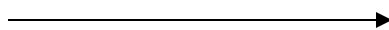
Plastics

- Polyethylene, high density (HDPE)



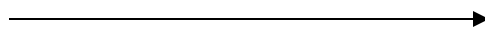
**Bottles, drums,
pipes, sheet, film,
etc.**

- Polypropylene, isotactic



**Automobile and
appliance parts, rope,
carpeting**

- Polystyrene, syndiotactic



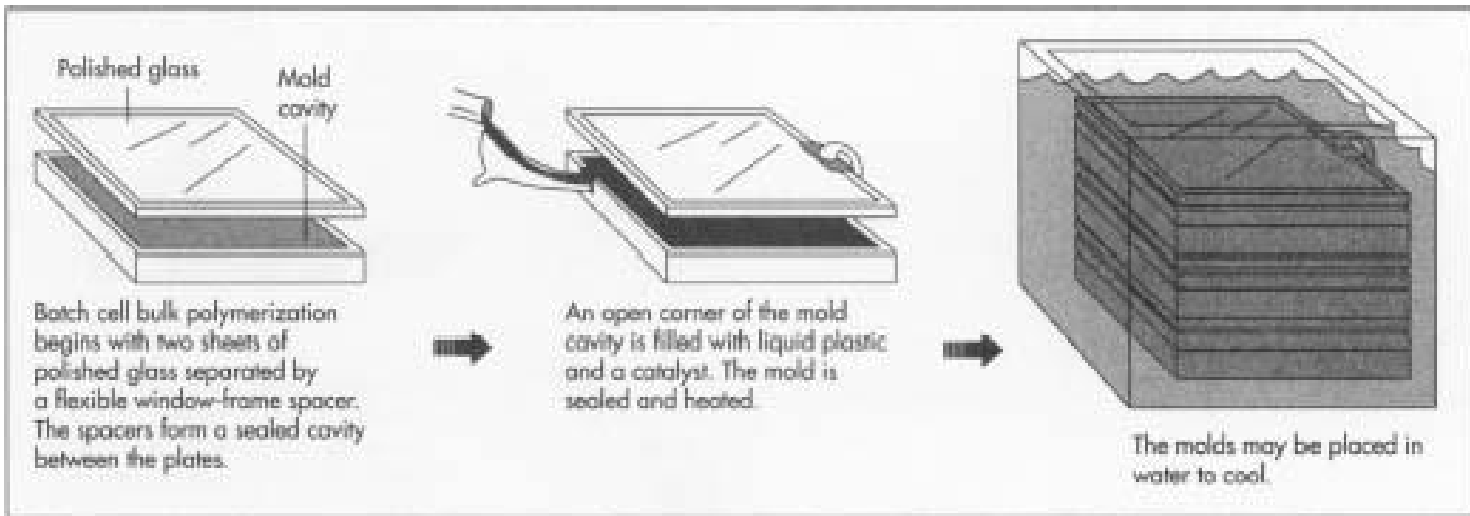
**Specialty
plastics**

Polymerization Techniques

- Bulk (no solvent just monomer)
- Solution
- Suspension
- Emulsion

Bulk Polymerizations

Acrylic sheets by Bulk polymerization of MMA



Rare

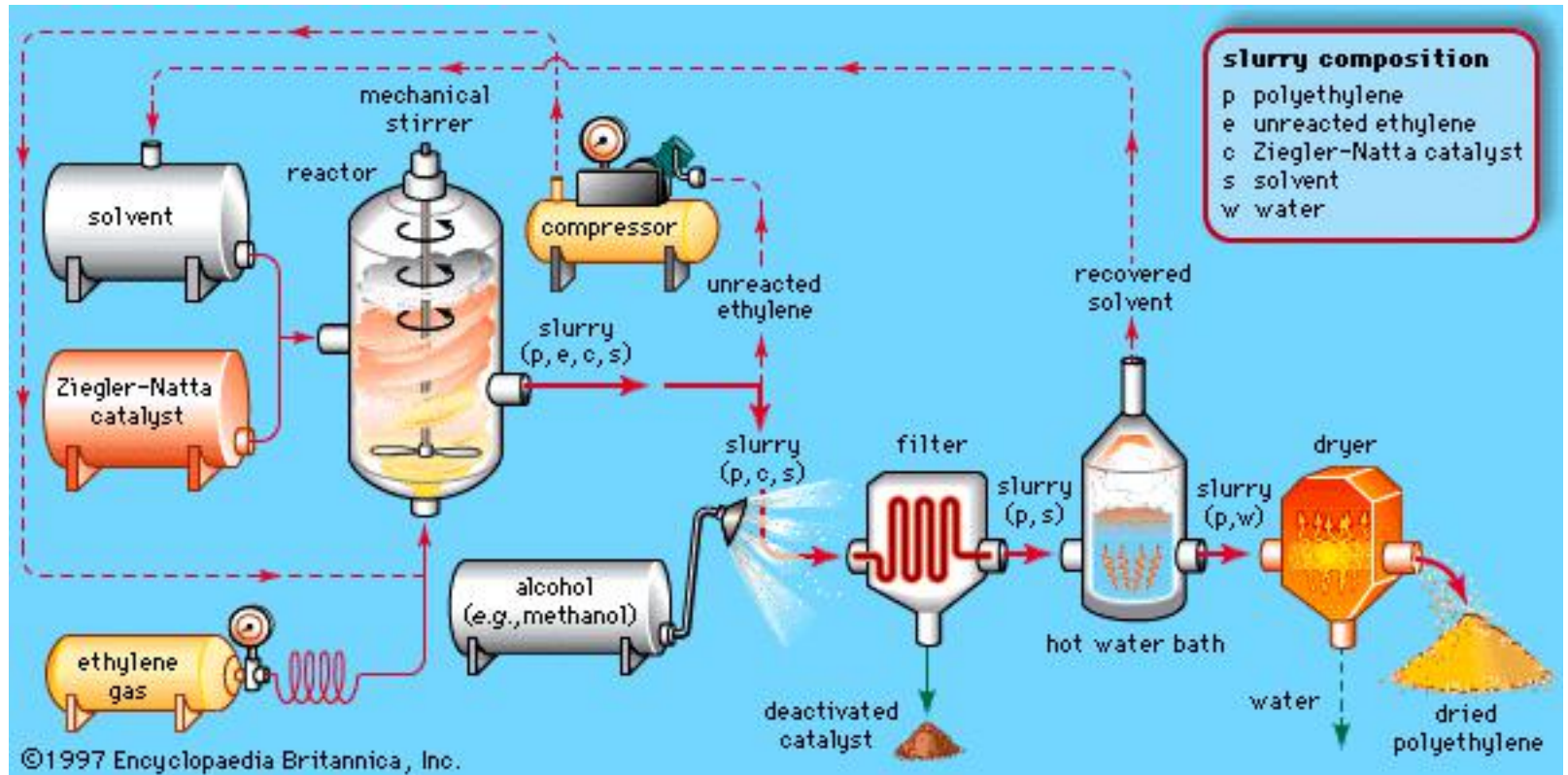
Overheat & explode with scale up

No solvent-just monomer

Polymer usually vitrifies before done

Broad MW distribution

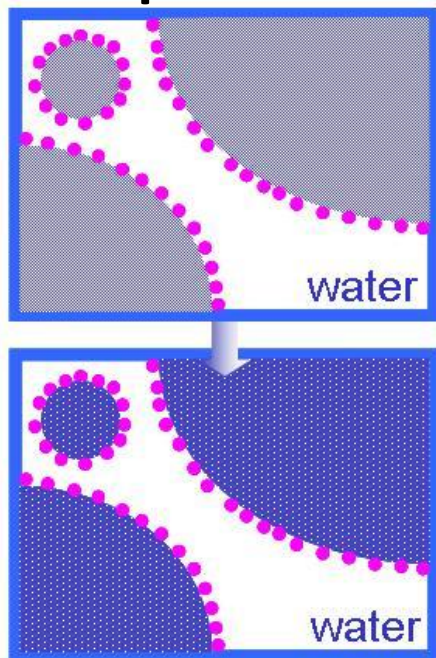
Solution Polymerization



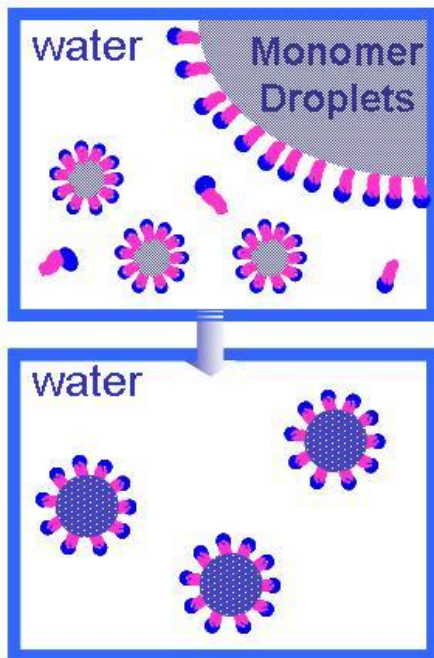
- Better control of reaction temperature
- Better control of polymerization
- Slower
- Not very green

Suspension vs. Emulsion

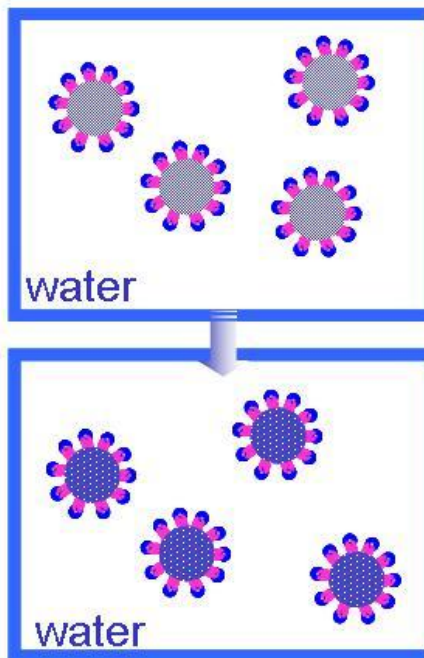
Suspension



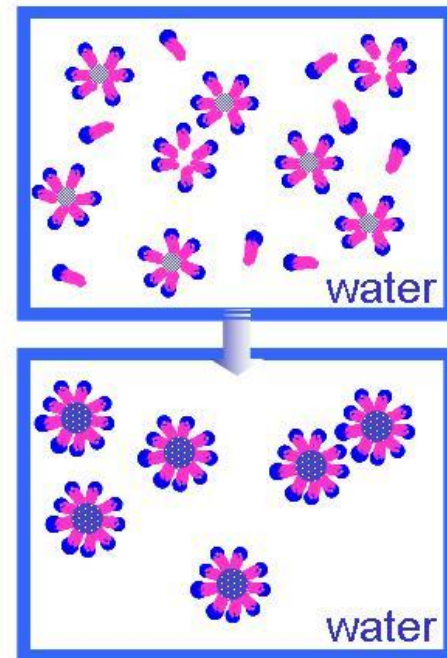
Emulsion



Mini-emulsion



Micro-emulsion



Monomer in oil

Monomer in oil

Monomer in oil

Monomer in oil

Initiator in oil

Initiator in water

Initiator in water

Initiator in water

Suspension Polymerization

- Oil droplets dispersed in water
- Initiator soluble in oil
- Greener than solution polymerization

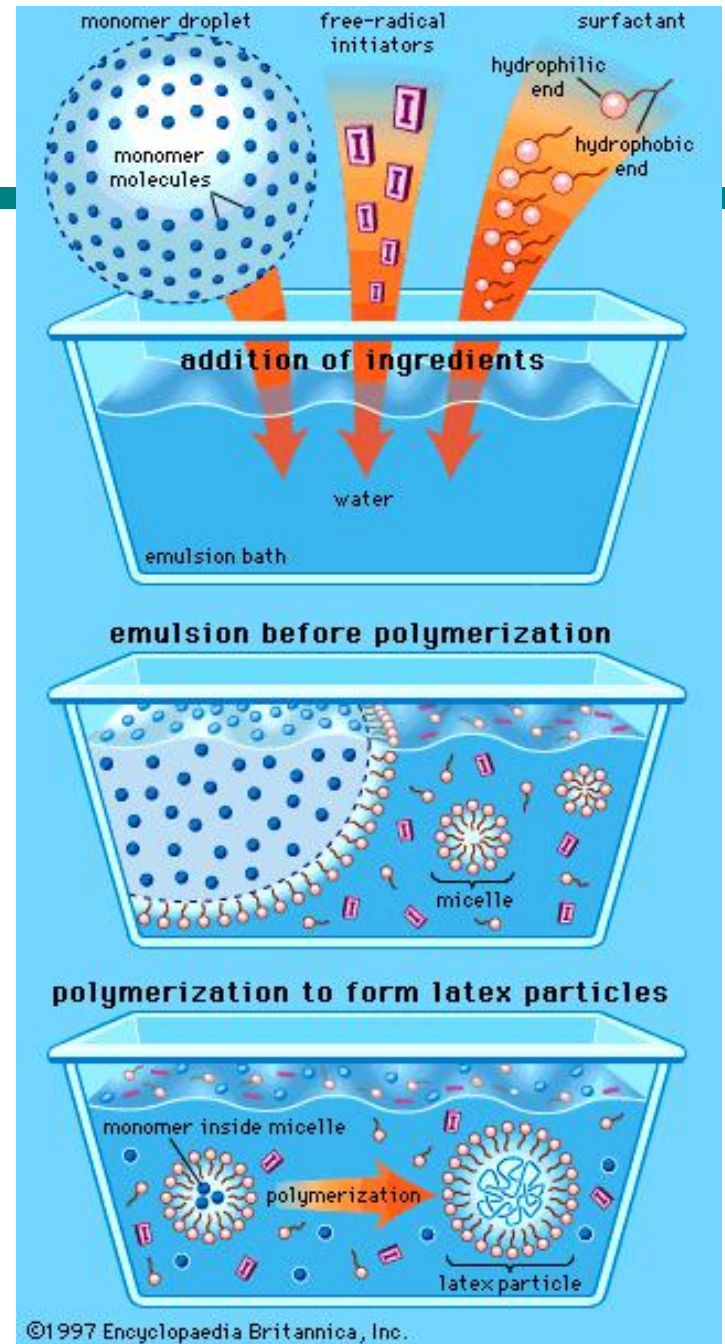


Filter off particles of polymer

Emulsion Polymerization



Still oil in water (or the reverse)
Initiator in water
Smaller particles (latex)
Excellent control of temp
Solution turns white

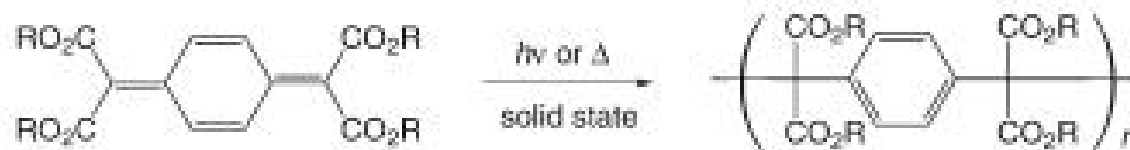


Less Common Polymerization Techniques

- Solid state polymerization
 - Polymerization of crystalline monomers
- Gas Phase polymerization
 - Parylene polymerizations
- Plasma polymerization
 - Put anything in a plasma

Solid State Polymerizations

Topological Polymerizations: Polymerization of crystals

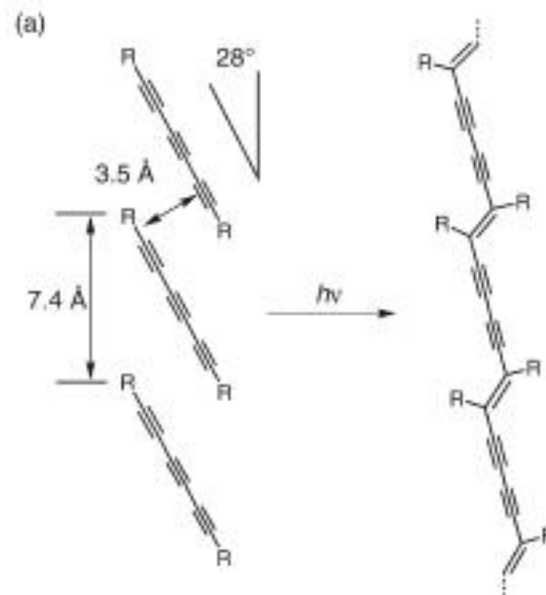


$\text{R} = \text{CH}_3, \text{CH}_2\text{CH}_2\text{Cl}, \text{CH}_2\text{CH}_2\text{Br}$

Scheme 11 Topochemical polymerization of 7,7,8,8-tetrakis(alkoxycarbonyl)quinodimethanes in the solid state under photoirradiation or upon heating

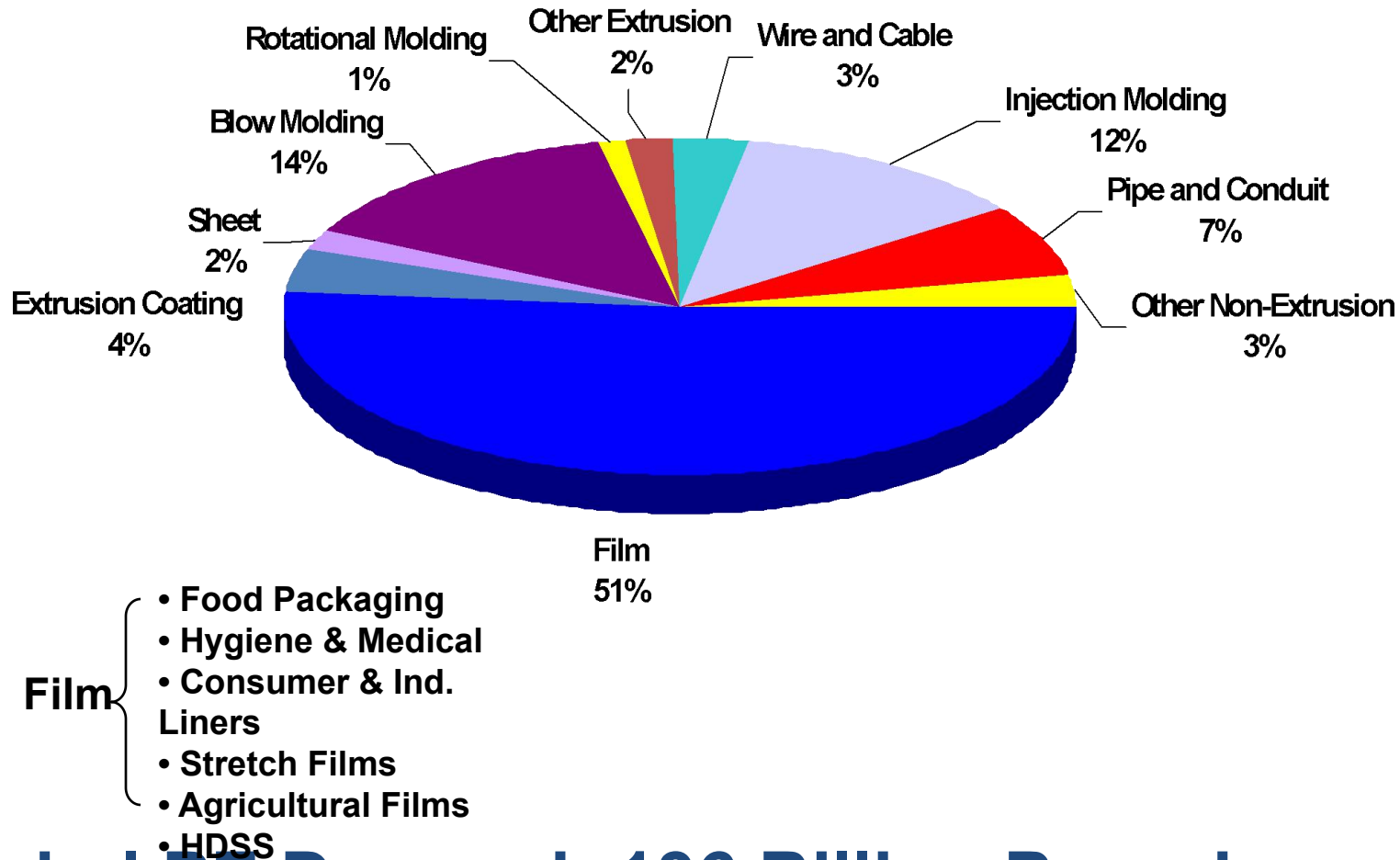
Quinodimethane polymerizations

Di- and Triacetylene polymerizations
In single crystals



Gas Phase Polymerization: LIGHT OLEFINS

Ethylene

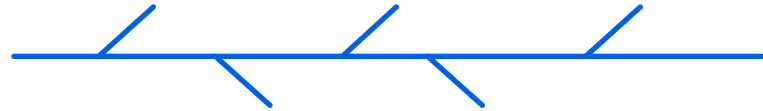


2004 Global PE Demand: 136 Billion Pounds

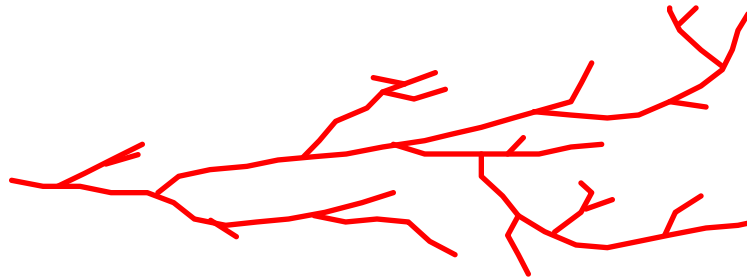
Types of Polyethylene



HDPE (0.940-0.965)
“High Density”



LLDPE (0.860-0.926)
“Linear Low Density”



LDPE (0.915-0.930)
“Low Density”

Gas Phase Polymerization: Light olefins

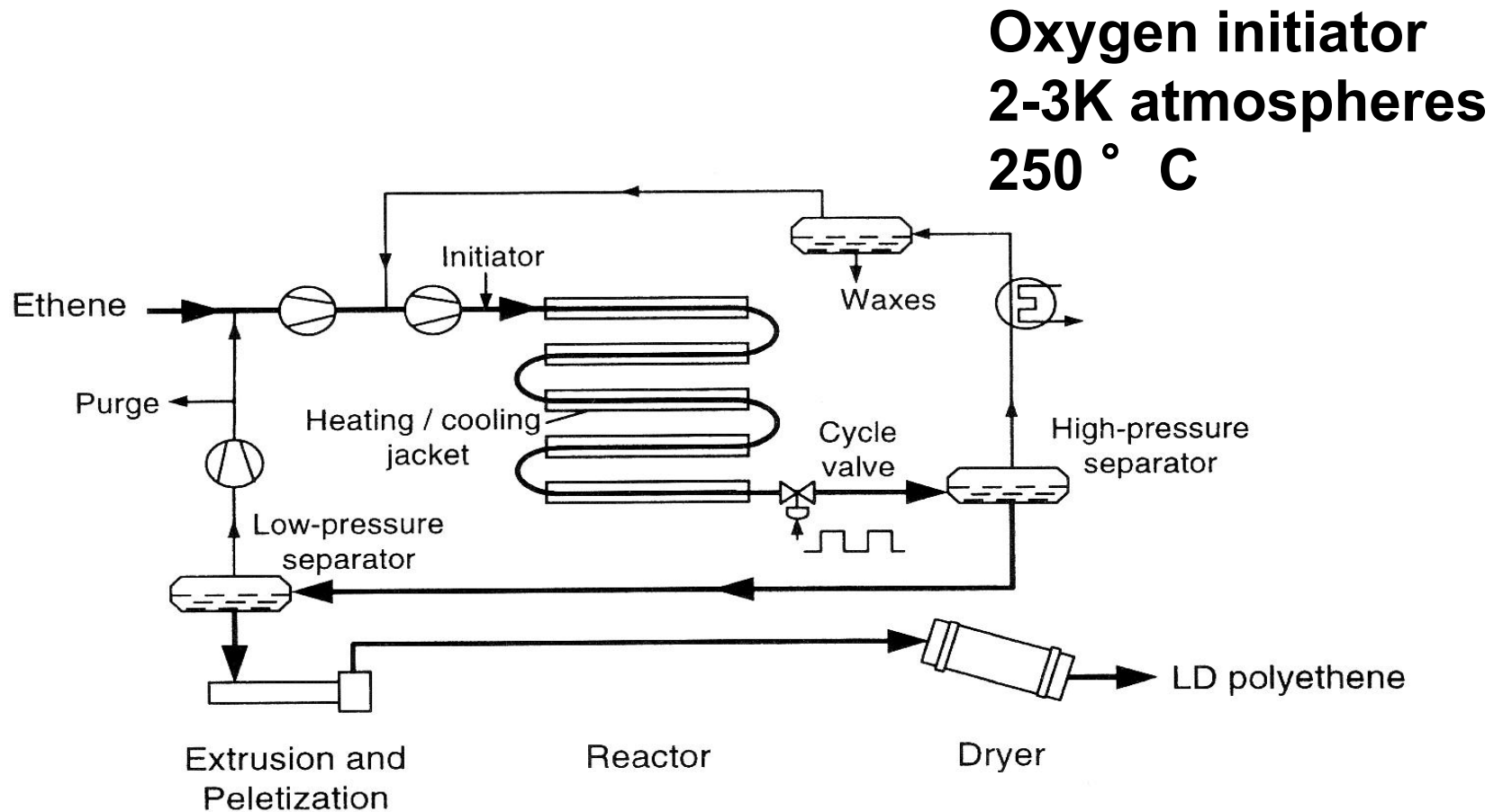


Figure 11.8 Production of LDPE in a tubular reactor (shown schematically).

Gas Phase Polymerization: Light olefins

Fluidized bed polymerization

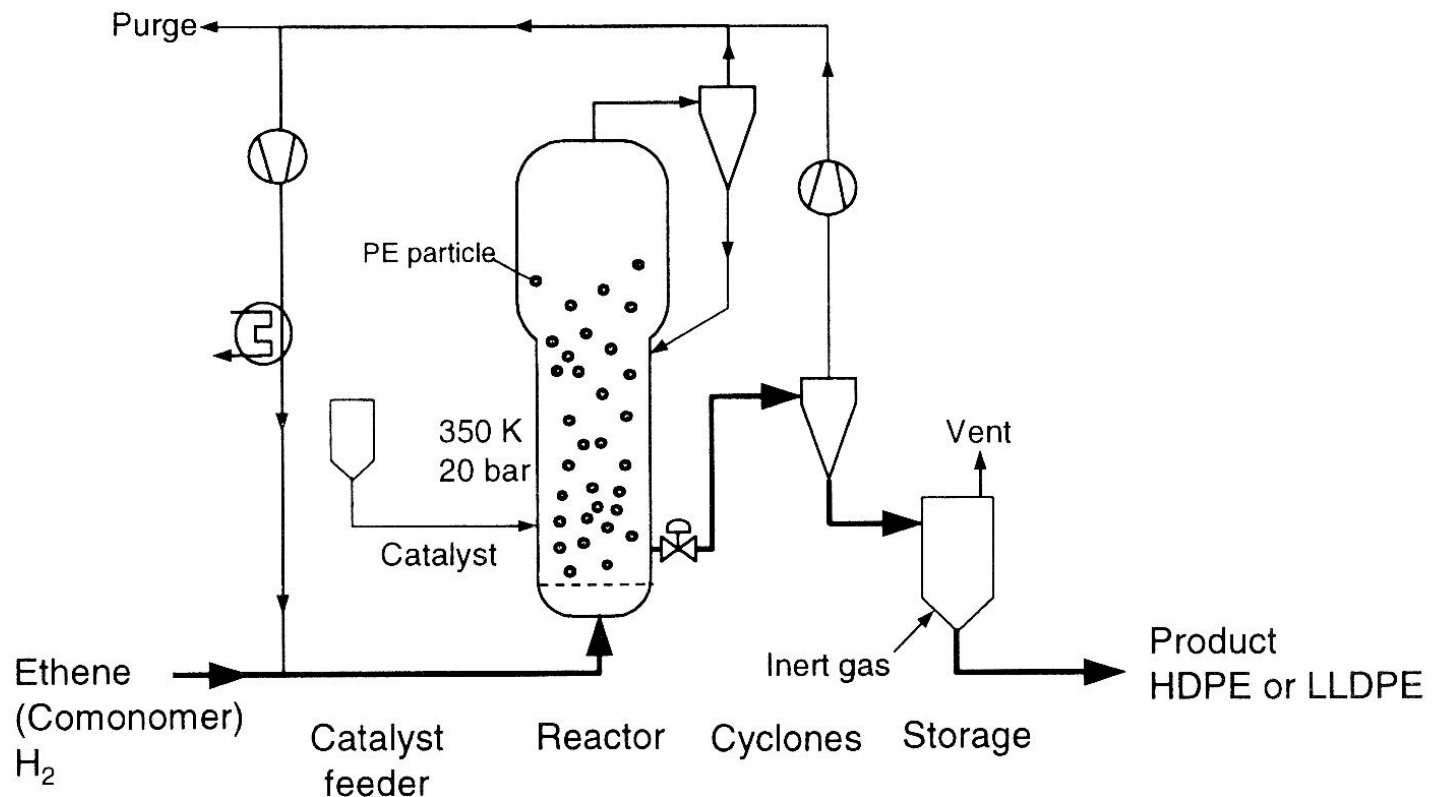
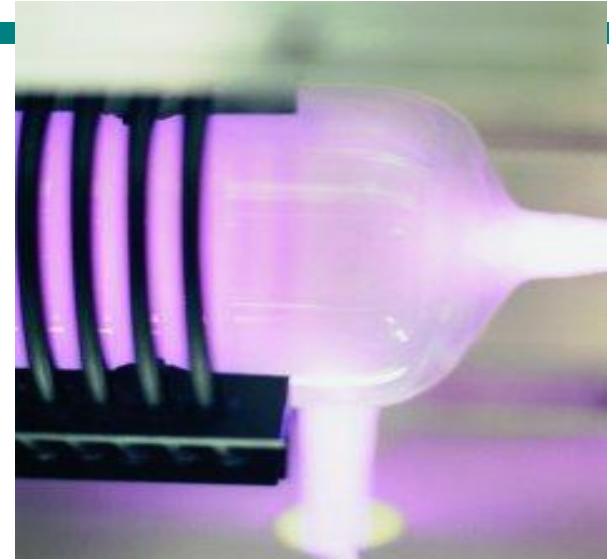
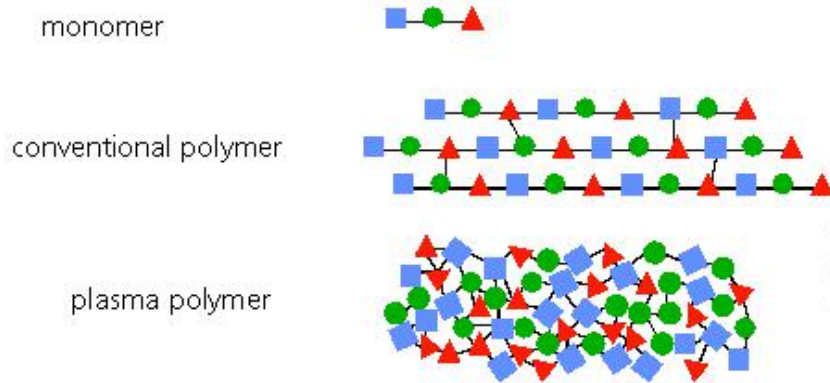


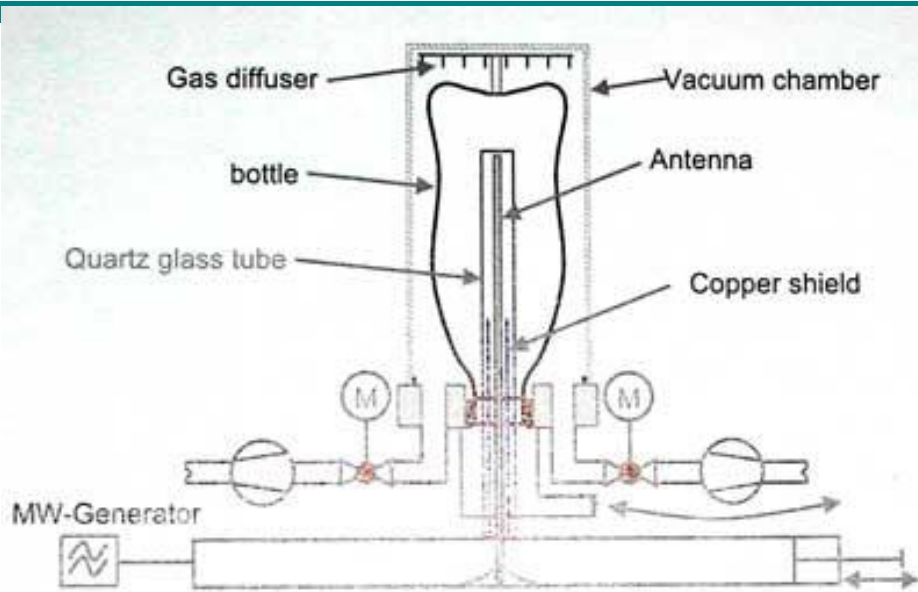
Figure 11.12 Union Carbide Unipol process for the production of HDPE and LLDPE.

Plasma Polymerization

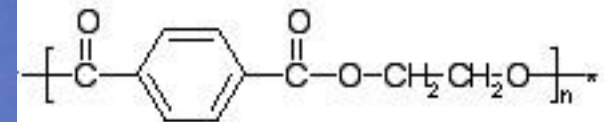


- 50 nm - 1 micron thick films
- Continuous coatings
- Solvent free
- High cohesion to surface
- Highly cross-linked
- Generally amorphous

Plasma Polymerization



Multi-layer bottles



PET [Poly(Ethylene Terephthalate)]

HOMEWORK

- Classify all the polymerizations listed in Page 12, and explain why you do so.
- Make a comparison of Chain Growth Polymerization with Step Growth Polymerization, as much as you can.